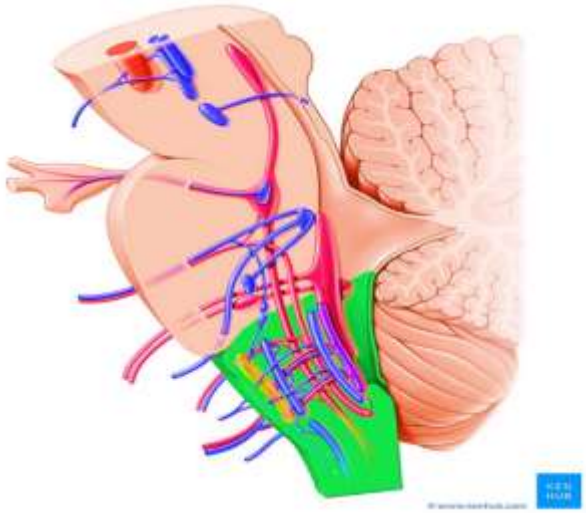
Medulla Oblongata

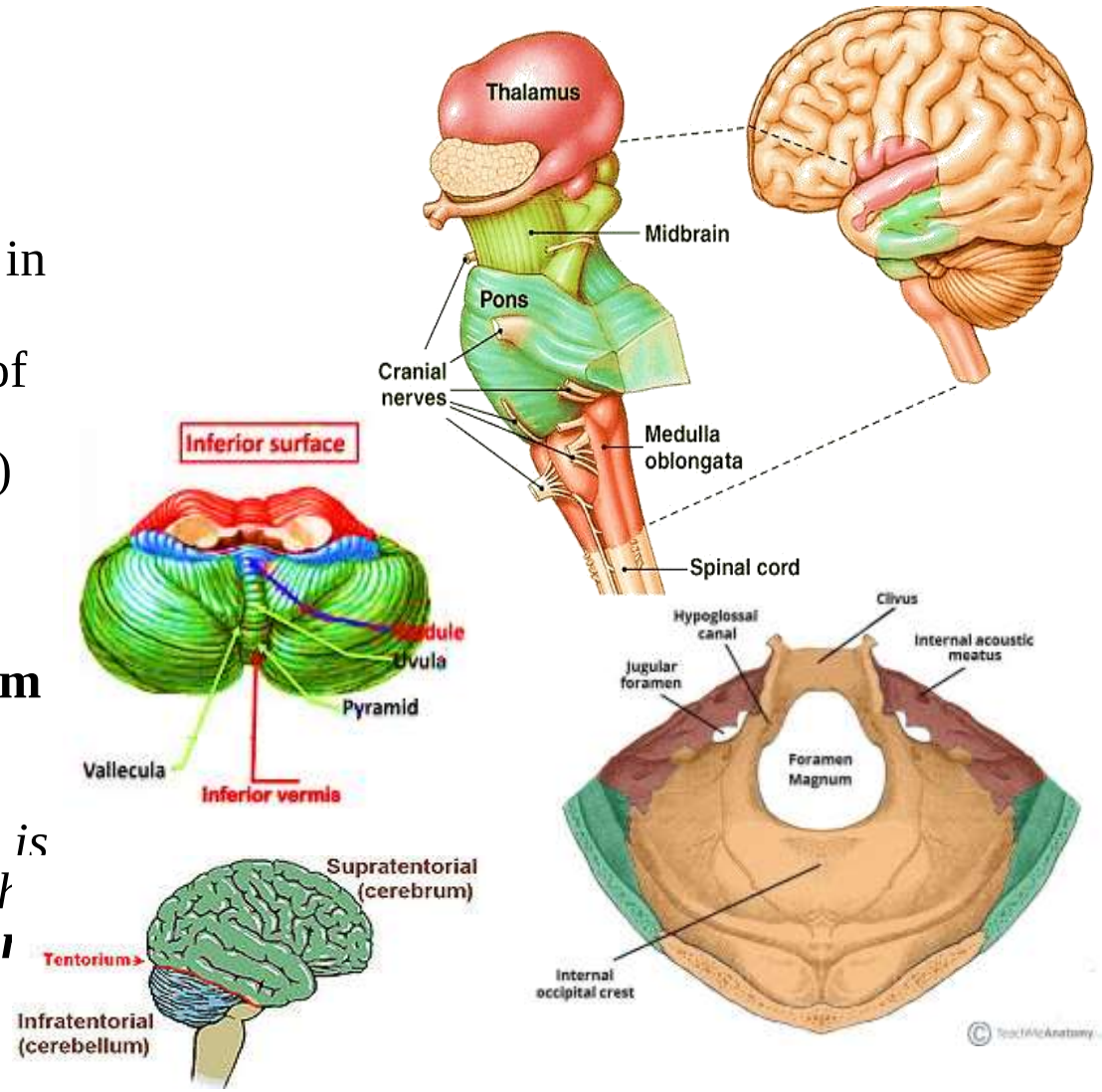


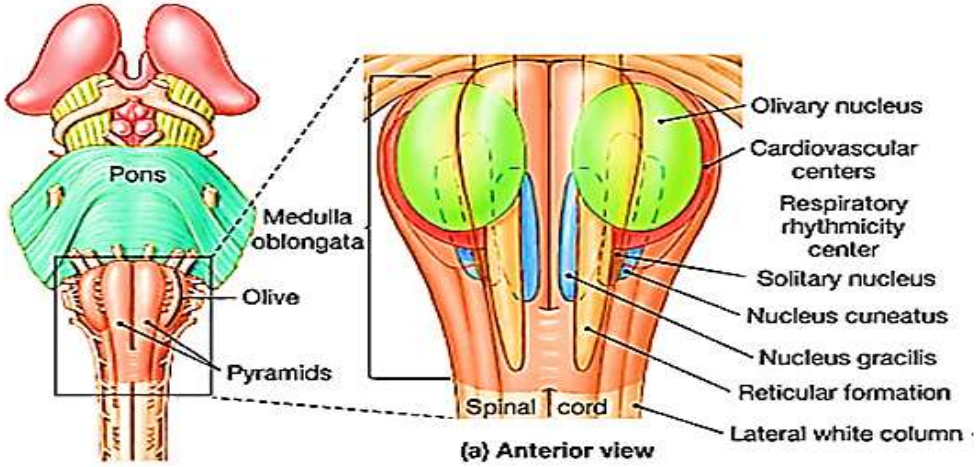
Outline

- **Overview & External features**
- **Sulci & Fissures of the medulla oblongata (MO)**
- **Nerves attached to the Surface of the Medulla**
- **Parts of the Medulla Oblongata**
- **Transverse sections of the MO at the level of the sensory decussation**
- **Transverse sections of the MO at the level of the pyramidal decussation.**
- **Applied Anatomy**

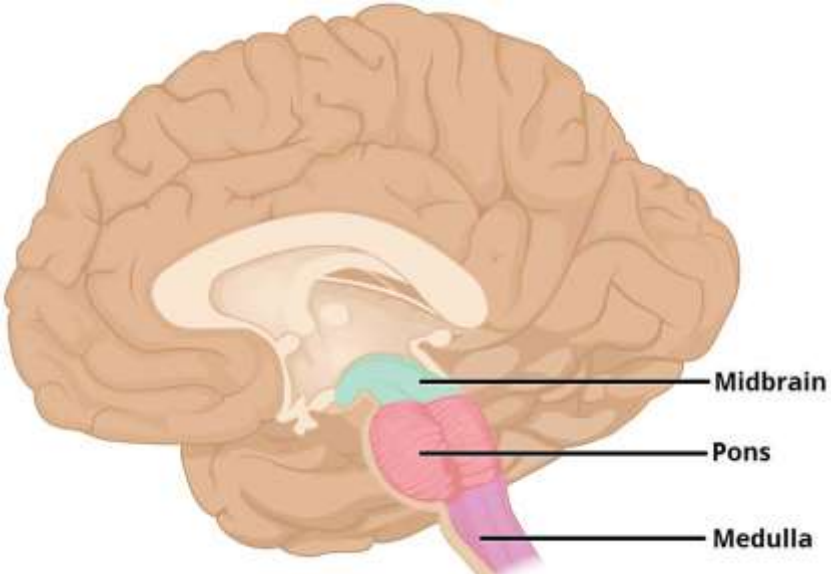
Overview & External Features

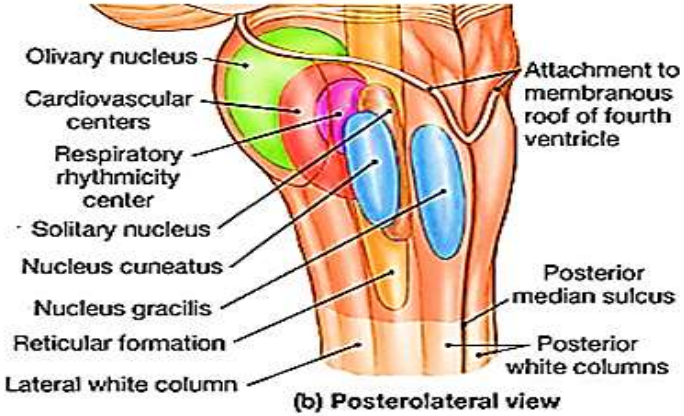
- The **medulla oblongata** is the **upward** prolongation of the **spinal cord** measuring **25 mm** in **length** & **18 mm** in **diameter**
- It is located in the **front area** of the **posterior cranial fossa** between the **clivus** (**anteriorly**) & the **vallecula**, groove or furrow of the **cerebellum** (**posteriorly**).
- It is connected to the **cerebellum** via the **inferior cerebellar peduncle**.
- *Note: **clivus** (Latin for "slope") is a bony part of the **cranium** at th skull base, a **shallow depressio** behind the **dorsum sellæ** that slopes obliquely backward.*



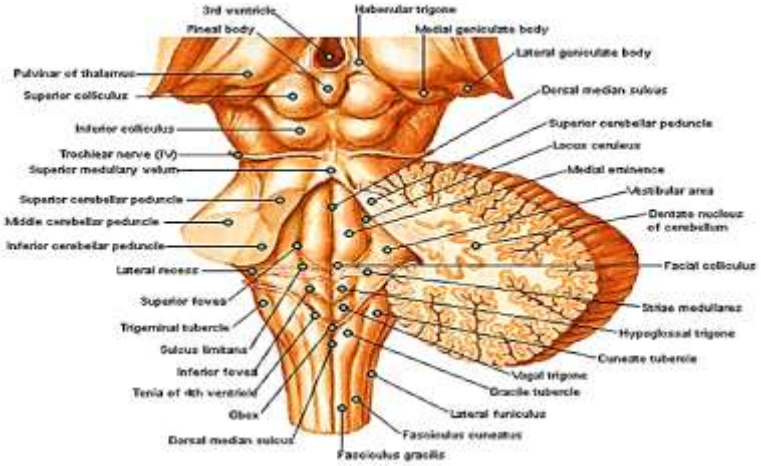


(a) Anterior view

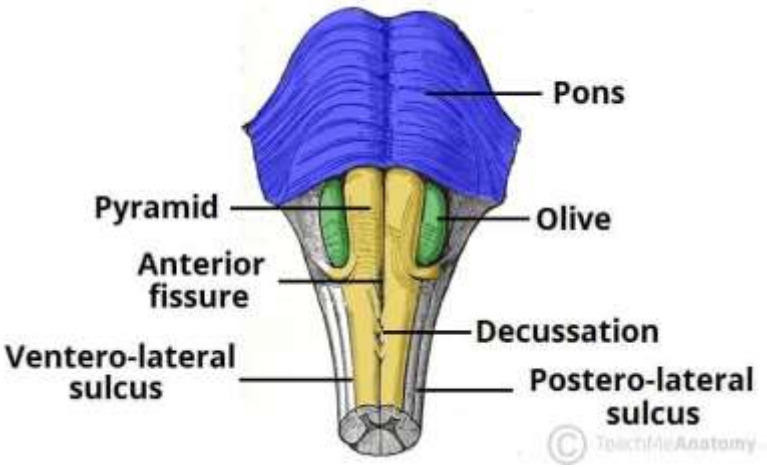




(b) Posterolateral view



- **Medulla oblongata** extends from the **lower border** of the **pons** at the **inferior pontine sulcus** spreading out to the **upper border** of the **atlas vertebra** opposite the **middle** of the **dens** at the **pyramidal decussation**.
- It contains **autonomic centres** that regulate:
  - (a) respiration,
  - (b) circulation &
  - (c) gastrointestinal motility.
- It gives rise to **9th, 10th, 11th** & **12th CNs,** the **5th** & **8th nuclei** extend **caudally** into the **medulla**.



MO lies beneath the tentorium cerebelli & above the foramen magnum in the posterior cranial fossa, it is related anteriorly to the basal portion of the occipital bone & the upper part of the odontoid process of the axis & posteriorly to the cerebellum.

- **Sulci & Fissures of the medulla**

- There is the presence of an **anterio-lateral sulcus** & a **postero-lateral sulcus**
- The anterior surface shows an **anterior median fissure**
- The posterior surface shows a **posterior median fissure**

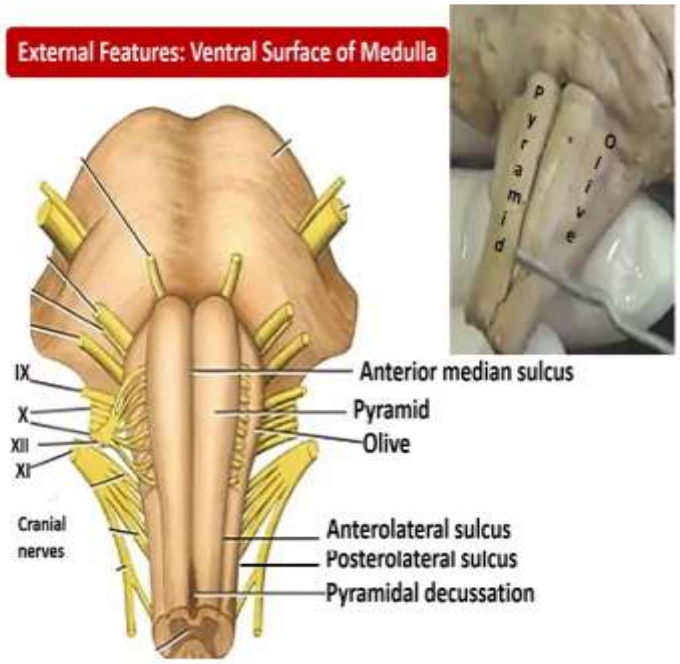
- **1. Elevations on the surface of the medulla:**

(a) Pyramids:

- It is a **longitudinal pyramidal elevation**, one on each side of the anterior median fissure
- Constitute the base of the medulla.
- Contain **uncrossed corticospinal fibers**.

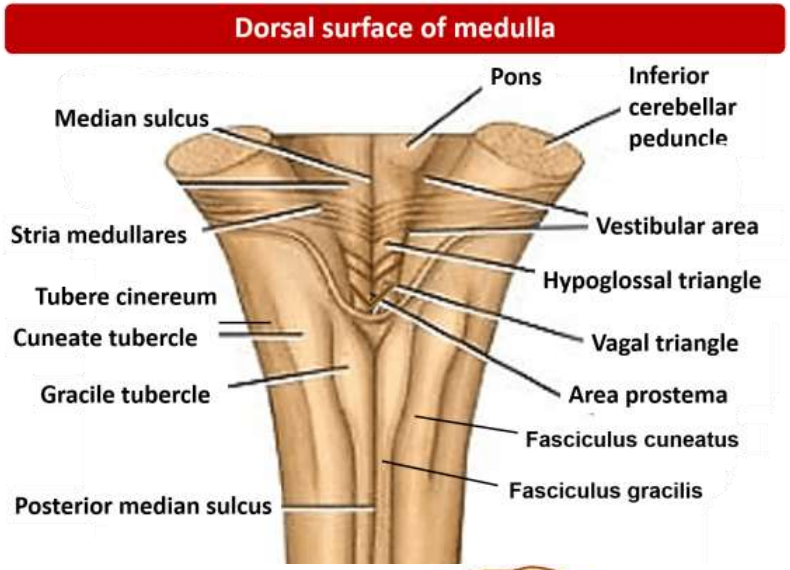
(b) Olive:

- It is an **oval elevation** (like an olive) just lateral to the **upper part** of the **pyramid**; it is formed by the **inferior olivary nucleus** which is a part of the **extrapyramidal system**.



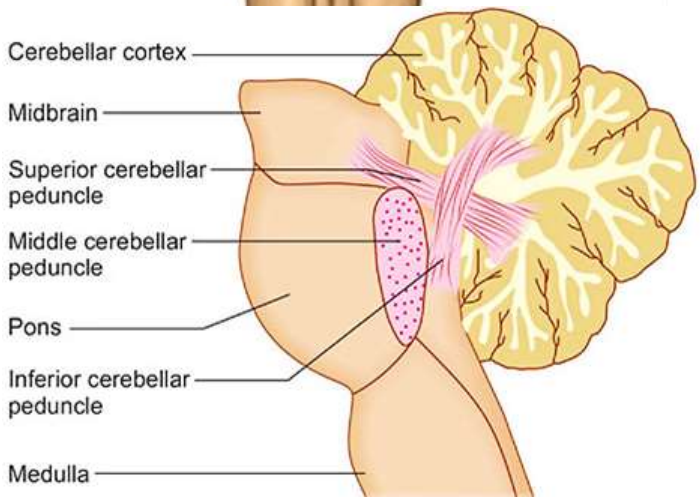
(c) Gracile & Cuneate Tubercles:

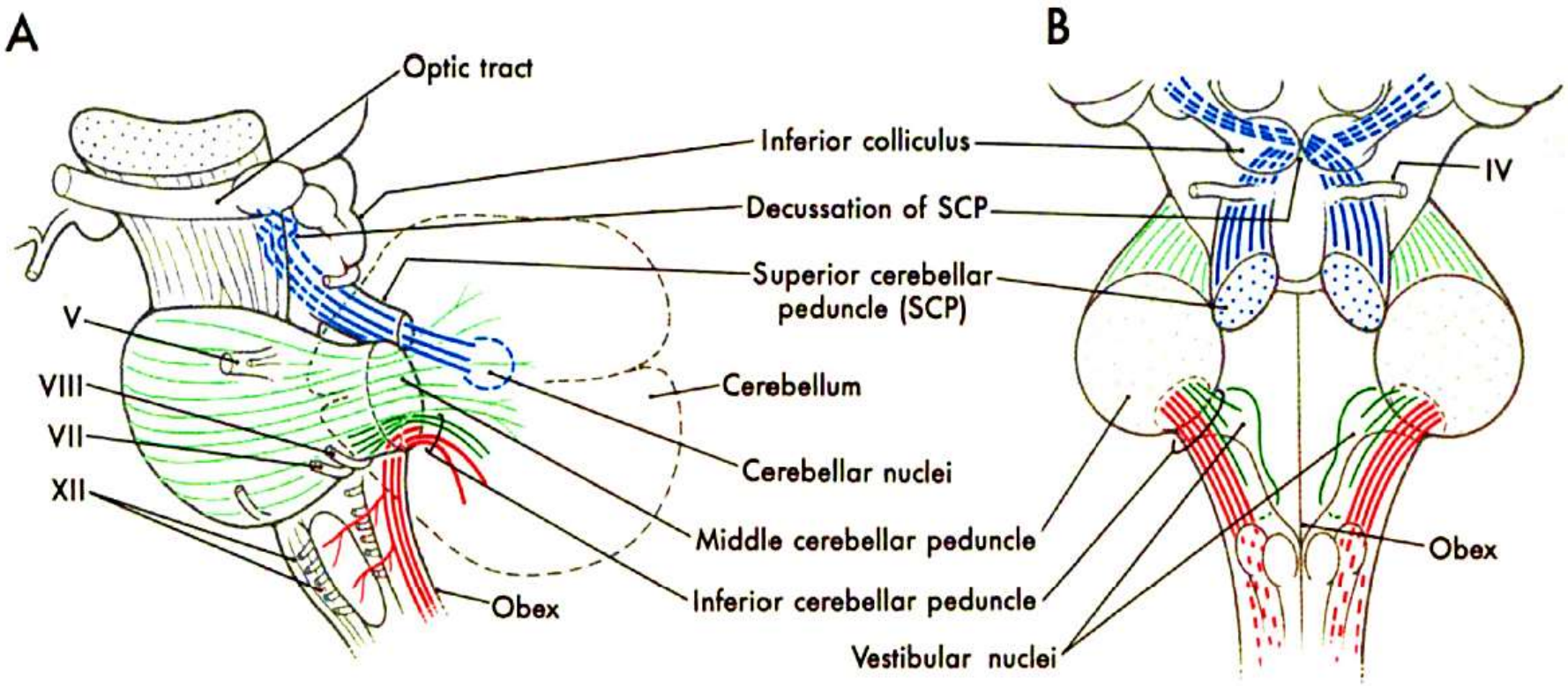
- These are **two elevations** on the **posterior** surface of the medulla, produced by the **gracile** & **cuneate nuclei**, the cuneate being lateral to the gracile. They are the upper ends continuation of the gracile & cuneate fascicules. Lateral to the tubercles is the **restiform body**, a prominent elevation on the posterolateral aspect of the medulla



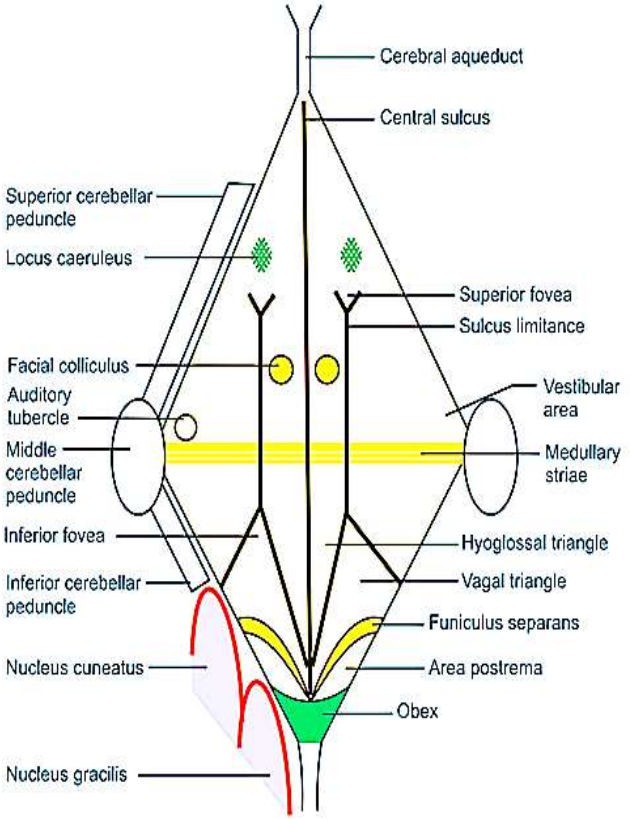
(d) Inferior cerebellar peduncle:

- It is an **elongated elevation** which lies posterolateral to the olive. The fibers connect the medulla as well as the spinal cord to the cerebellum. The inferior **cerebellar peduncle comprises** the **juxtarestiform body** & **restiform body**

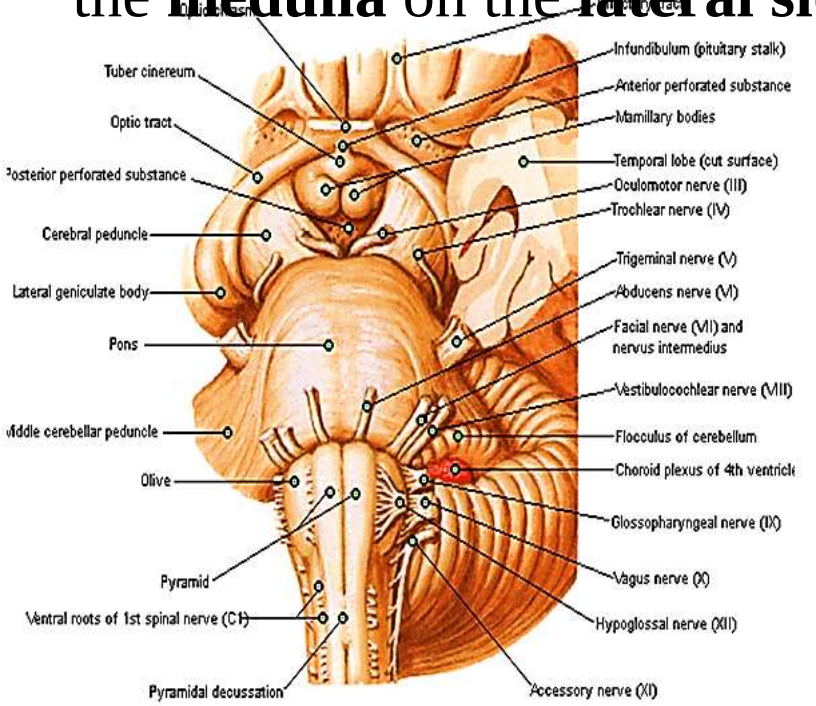


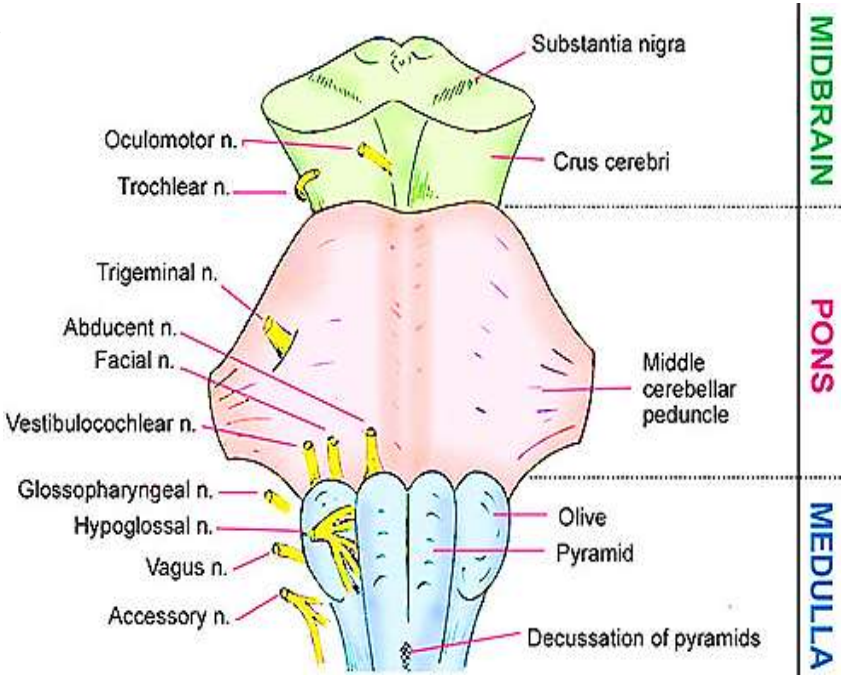

A
Optic tract
Inferior colliculus
Decussation of SCP
Superior cerebellar peduncle (SCP)
V
VIII
VII
XII
Cerebellum
Cerebellar nuclei
Middle cerebellar peduncle
Inferior cerebellar peduncle
Obex
B
IV
Obex
Vestibular nuclei

- **2. Depressions:**
- On the **upper part** of the **posterior surface** (**lower part** of the **floor** of the **4th ventricle**) in the **floor** of the **4th ventricle** & above the **obex**, is a **tongue-shaped depression**, the **area postrema**. The **area postrema** is a **circumventricular organ** with **capillaries** are **permeable** (i.e. no **blood–brain barrier**) & sensory neurons that enable it detect circulating **chemical messengers** in the **blood**.
- The **obex** (***Latin for barrier***) is the **point** at which the **4th ventricle narrows** to become the **central canal** of the **spinal cord**. The obex occurs in the **caudal medulla**, at which point the **decussation** of **sensory fibers** happens.

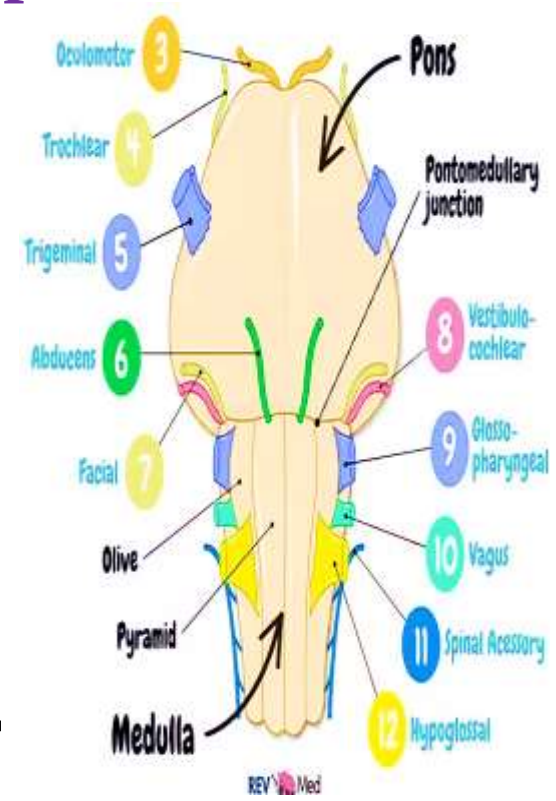


- **Nerves Attached to the Surface of the Medulla:**
- **At the junction between the pons & medulla**
  - **6th CN (Abducens nerve):** lies in line with rootlets of the **hypoglossal nerve** (near the anterior median fissure/anterolateral sulcus).
  - **7th CN (Facial nerve):** lies **posterolateral** to the **6th nerve**, in line with the **9th**, **10th** & **11th nerves**.
  - **8th CN (Vestibulocochlear nerve):** lies just lateral to the **7th nerve** & both of them emerge at the **cerebellopontine angle** (the angle between the **lower border** of the **pons** & **upper border** of the **medulla** on the **lateral side**).



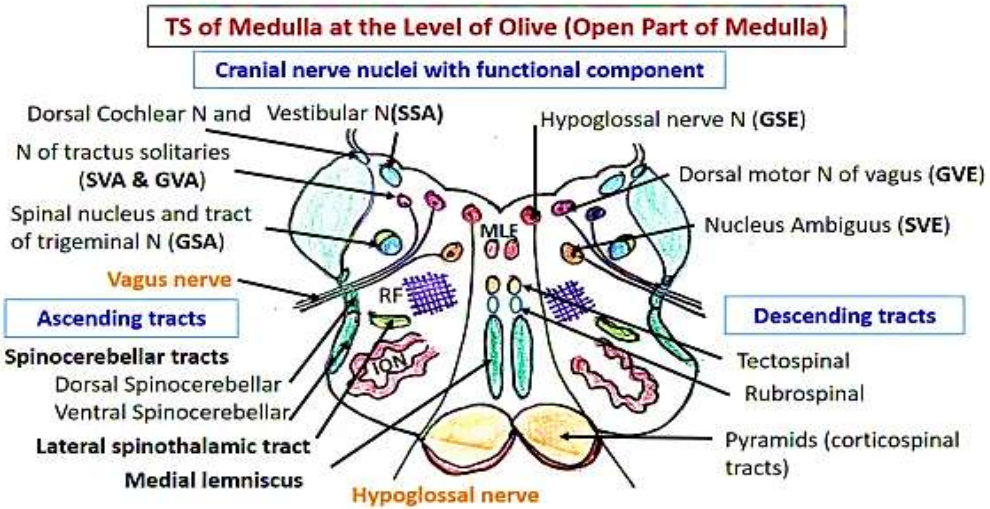


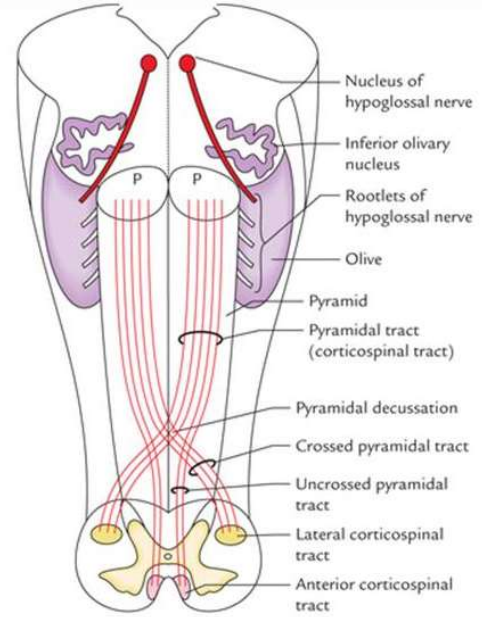
- **Between the olive & inferior cerebellar peduncle:**
- There are **3 nerves** arranged from **above downwards**
  - **9th CN (Glossopharyngeal nerve)**
  - **10th CN (Vagus nerve)**
  - **11th CN (Accessory nerve)**
- These nerves are attached to the **side** of the **medulla** one above the other in line with the **dorsal roots** of spinal nerves. In the **posterolateral sulcus** where the **three nerves** are attached lie the **spinal tract** & **spinal nucleus of trigeminal.** These nerves are in close relation with the **posterior inferior cerebellar artery** (**haemorrhage or aneurysm** from this **artery**, may affect these **three nerves**).
- **Between the olive & pyramid:**
- **12th CN (Hypoglossal nerve):** its rootlets lie in the line with the ventral roots of the spinal nerve. i.e., **anterolateral sulcus**.



Parts of the Medulla Oblongata

- The **upper part** of the medulla **(*open medulla*)** which forms part of the **4th ventricle** & has **no central canal**
- The **lower part** of the medulla **(*closed medulla*)** which has a **central canal** inside it.



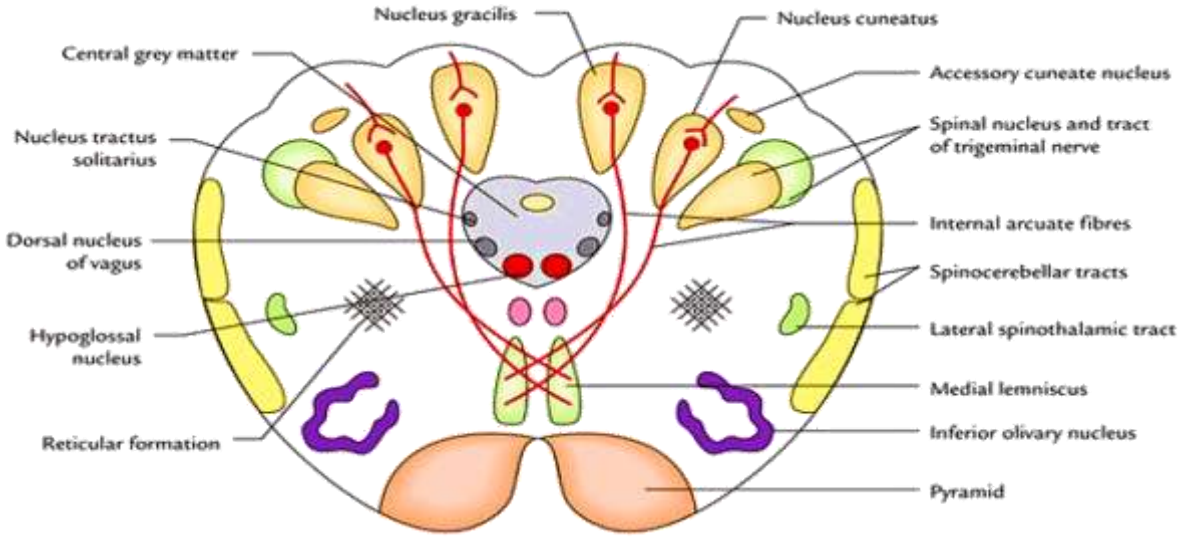


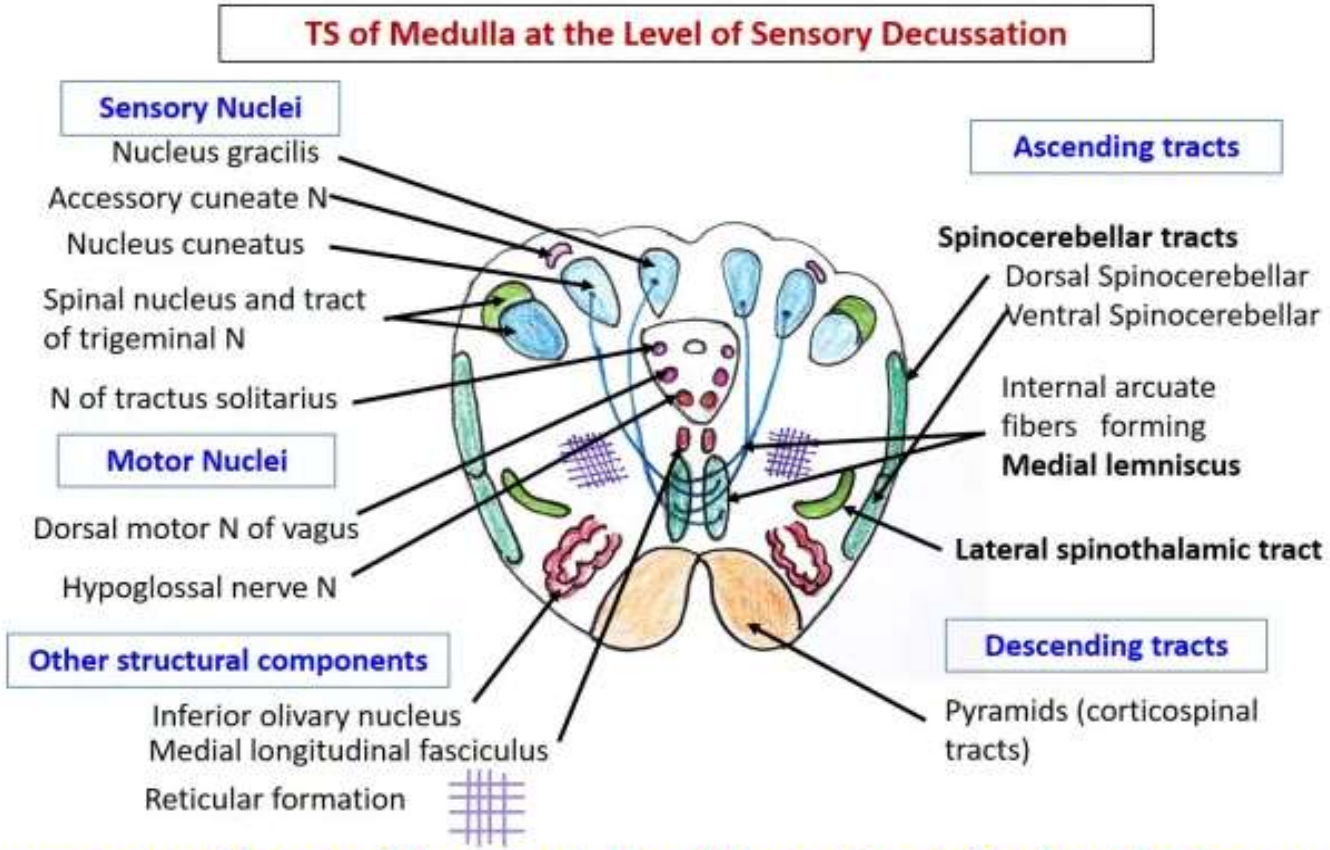
MLF- Medial longitudinal Fasciculus
ION- Inferior Olivary Nucleus
RF – Reticular Formation

Transverse sections of the medulla oblongata at the level of olive (open part of medulla)

Gray Matter of Medulla Oblongata

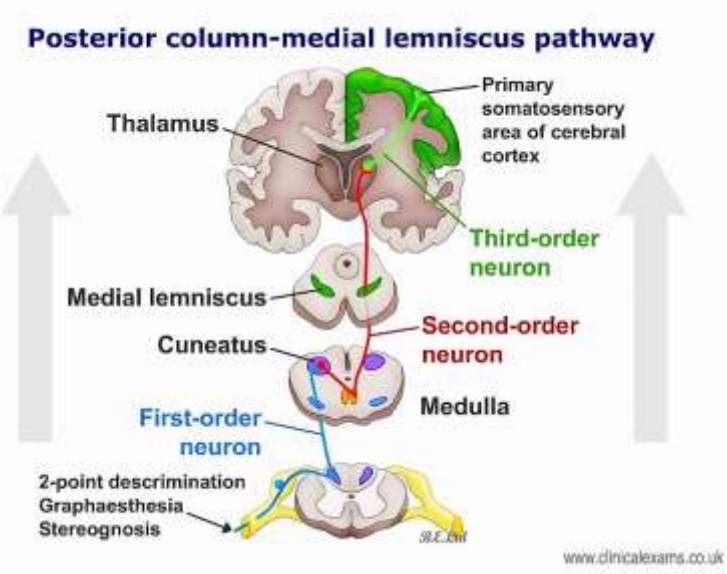
- **Gracile & Cuneate Nuclei:**
  - The **nuclei** lie in the **posterior** aspect of the **lower part** of **medulla** in line with the **gracile** & **cuneate tracts/fasciculus** whose fibers **synapse** with cells of such **nuclei**.
  - The **axons** of the **cells** of **gracile** & **cuneate nuclei (2nd order neurons)** are referred to as the **internal arcuate fibres** passes forward **decussate bending** to cross the **median plane** as the **medial lemnisci decussation** or **sensory decussation**.

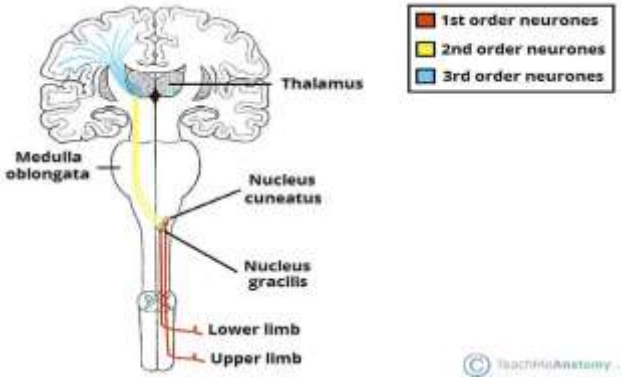




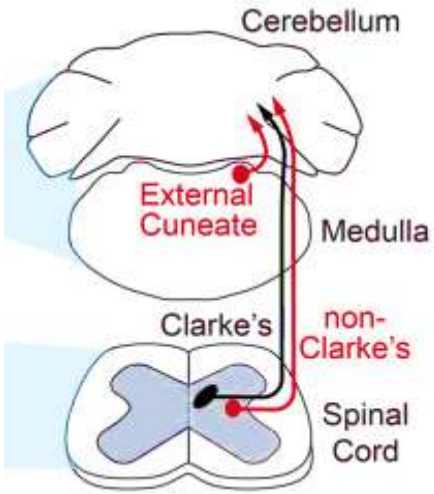
Transverse sections of the medulla oblongata at the level of sensory decussation

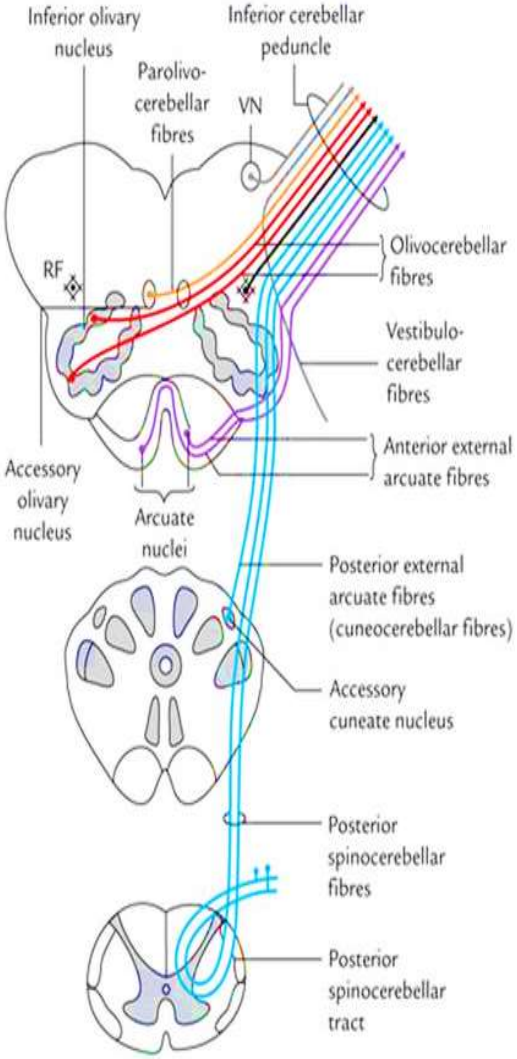
- They then **ascend** after decussating as a **compact bundle** called **medial lemniscus** [or **dorsal column** or **Reil's band**/ribbon (comprise **gracile** & **cuneate fasciculus**)].
- It ascends in the **medulla** close to the **median raphe**, **posterior** to the **pyramidal tract** & **anterior** to the **tectospinal tract** & **medial longitudinal bundle/ fasciculus**.
- The **medial lemniscus** is a pathway that carries **sensory information** (**proprioceptive** & **fine touch sensations**) from the **gracile** & **cuneate** nuclei to the **thalamus** **(ventral posterolateral nucleus)**.

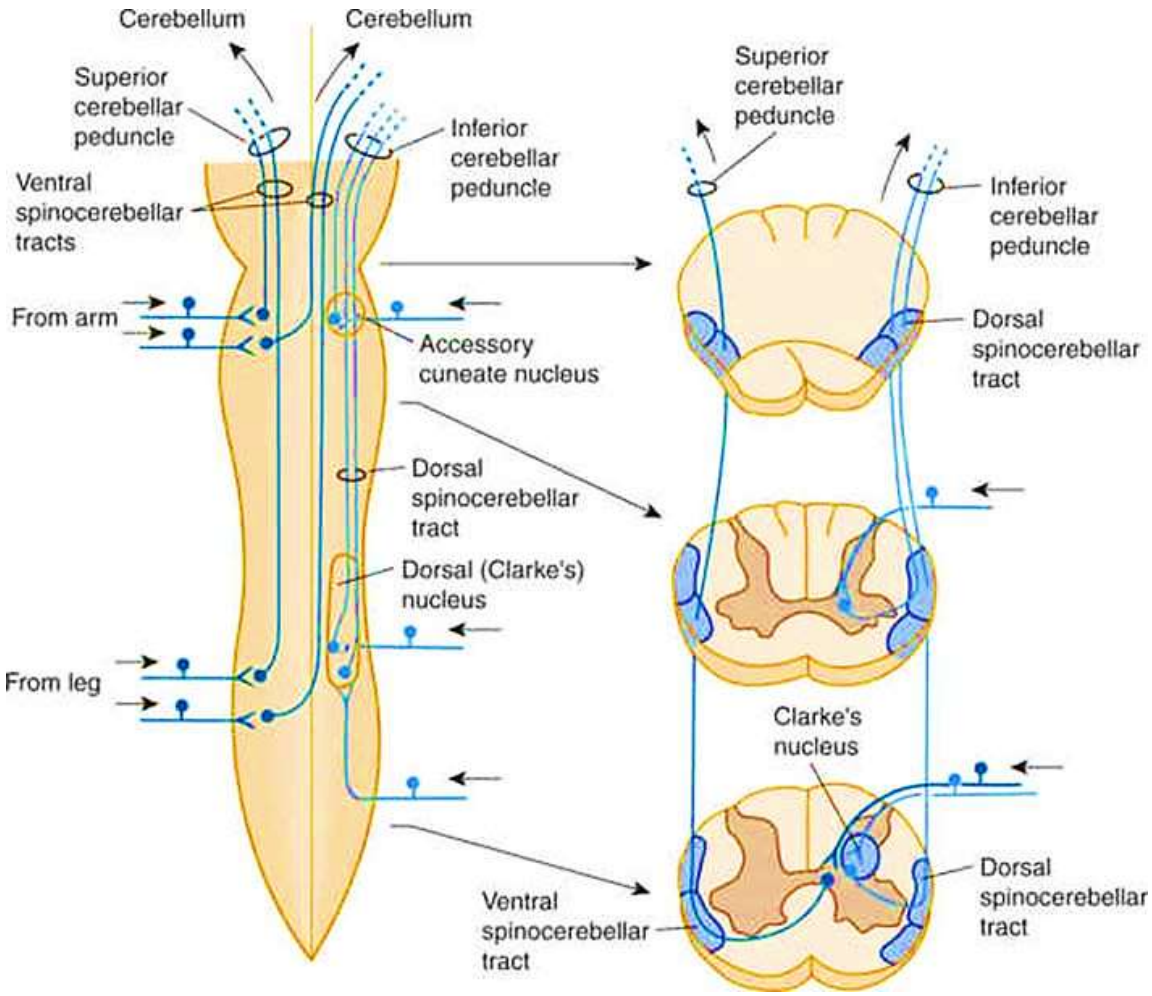




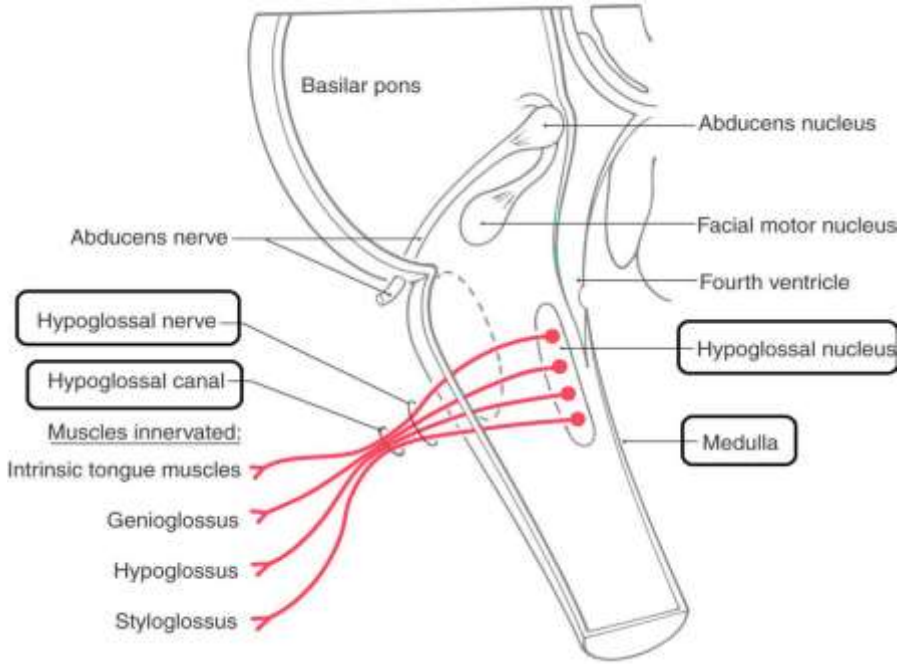
- **Accessory Cuneate Nucleus:**
  - This nucleus lies **posterolateral** to the **cuneate nucleus** in line with the **thoracic nucleus (Nucleus dorsalis of Clarke)**.
  - It **gives origin** to the **posterior external arcuate fibers** which convey **proprioceptive sensation** mainly from the **neck** & **upper limbs** to the **cerebellum** (to same side) via the **inferior cerebellar peduncle**.
  - Note that the **thoracic nucleus in the spinal cord** convey the same sensation as the **accessory cuneate nucleus**, but from the **lower limb** to the **cerebellum** (same side).

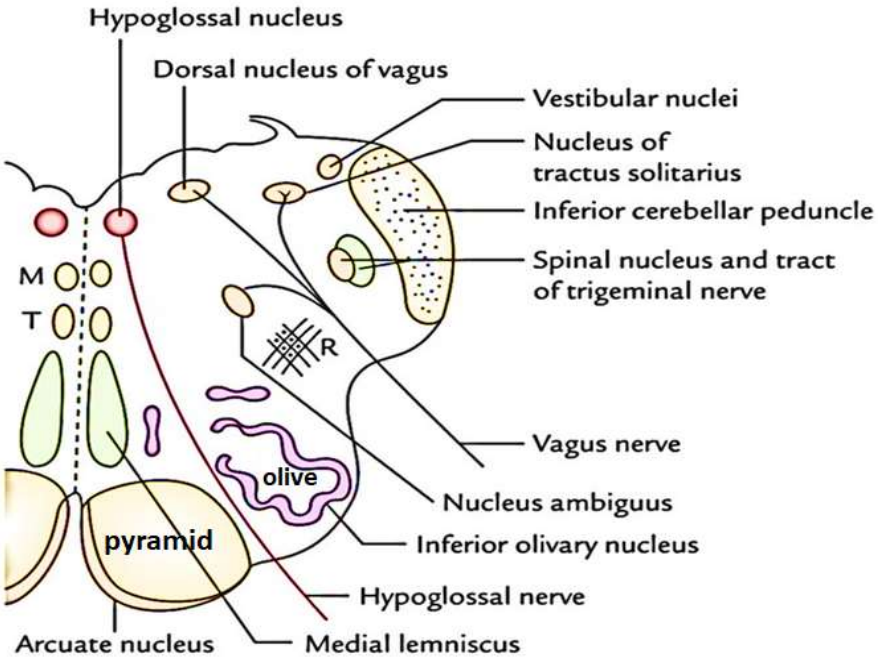




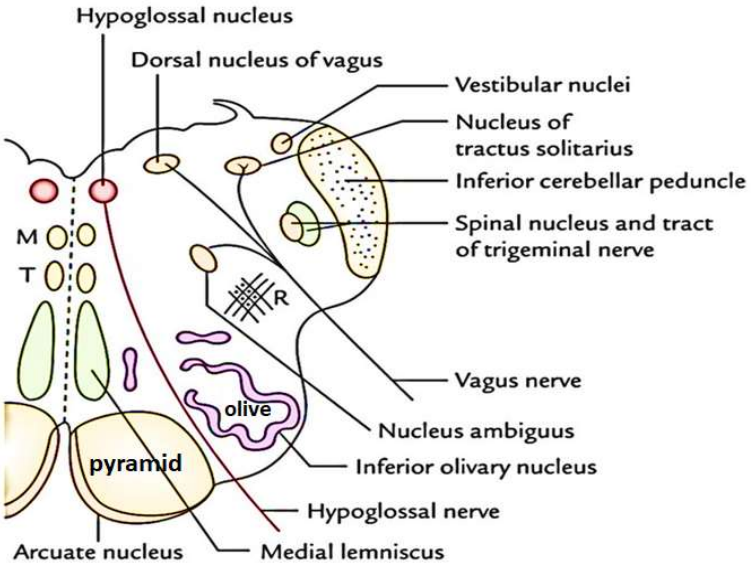

Cerebellum
Cerebellum
Superior cerebellar peduncle
Inferior cerebellar peduncle
Ventral spinocerebellar tracts
From arm
Accessory cuneate nucleus
Dorsal spinocerebellar tract
Dorsal (Clarke's) nucleus
From leg
Ventral spinocerebellar tract
Superior cerebellar peduncle
Inferior cerebellar peduncle
Dorsal spinocerebellar tract
Clarke's nucleus
Dorsal spinocerebellar tract

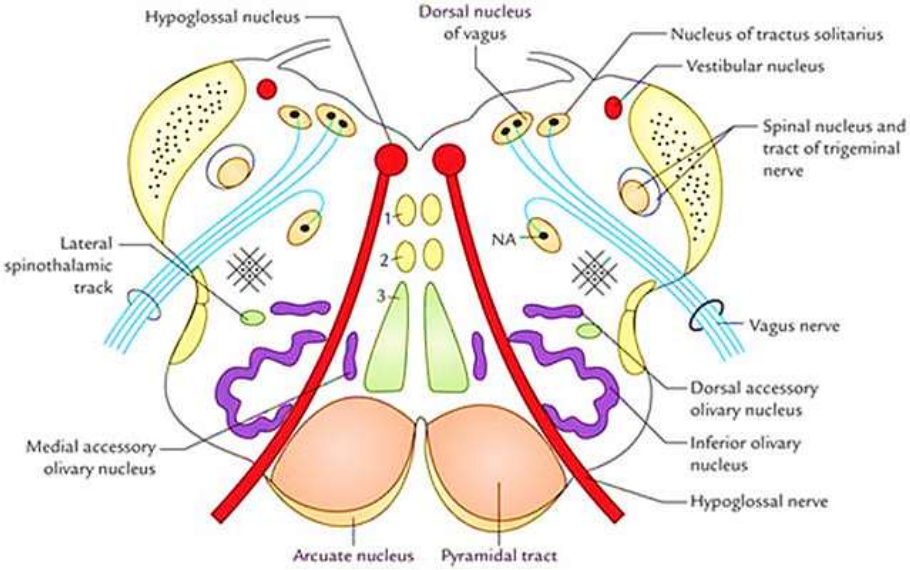
- **Hypoglossal Nucleus:**
  - The **hypoglossal nucleus** lies in the **floor** of the **lower part** of the **4th ventricle** close to the **midline**, it gives rise to **general somatic efferent** (**GSE**) fibers that innervate the **intrinsic** & **extrinsic muscles** of the **tongue**.
  - The fibers of the **hypoglossal nerve** travels frontwards **lateral** to the **medial lemniscus**, penetrates the **medial part** of the **olivary nucleus** & finally emerges on the surface **between** the **pyramid** & the **olive**.



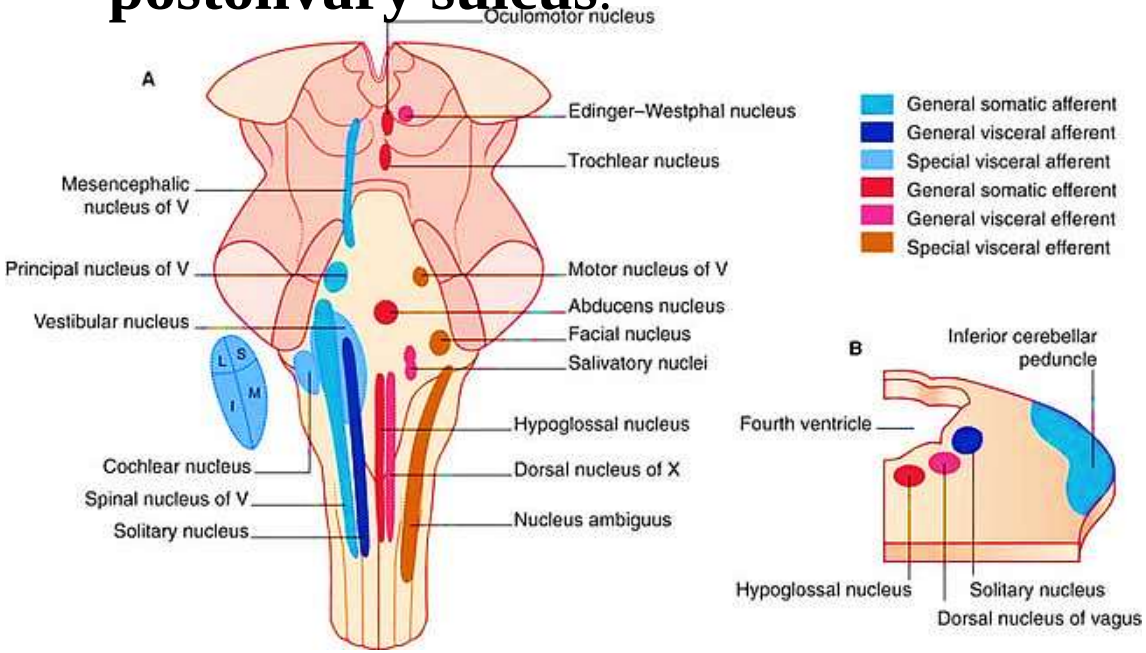


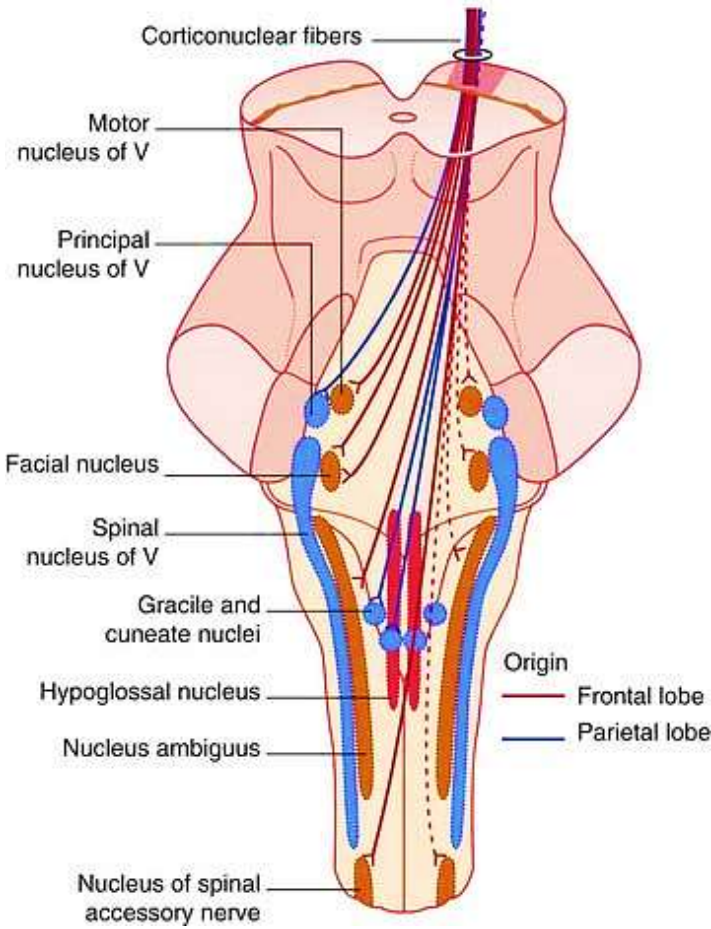
- **Dorsal Nucleus of Vagus:**
  - This **nucleus** lies just **lateral** to the **hypoglossal nucleus**.
  - The **cells** of this **nucleus** are mainly **parasympathetic** that send their **axons** to supply the **heart muscle**, **muscles of the respiratory tract** & **muscles of gut (general visceral efferent [GVE]).**
  - It is believed that some **cells** of this **nucleus** receive also **general visceral afferent (GVA)** impulses from the **same structures**.
  - Other significant **autonomic centers** such as the **vasomotor, respiratory**, **reflex centers for deglutition** & **vomiting** lie very close to the **dorsal nucleus of vagus**



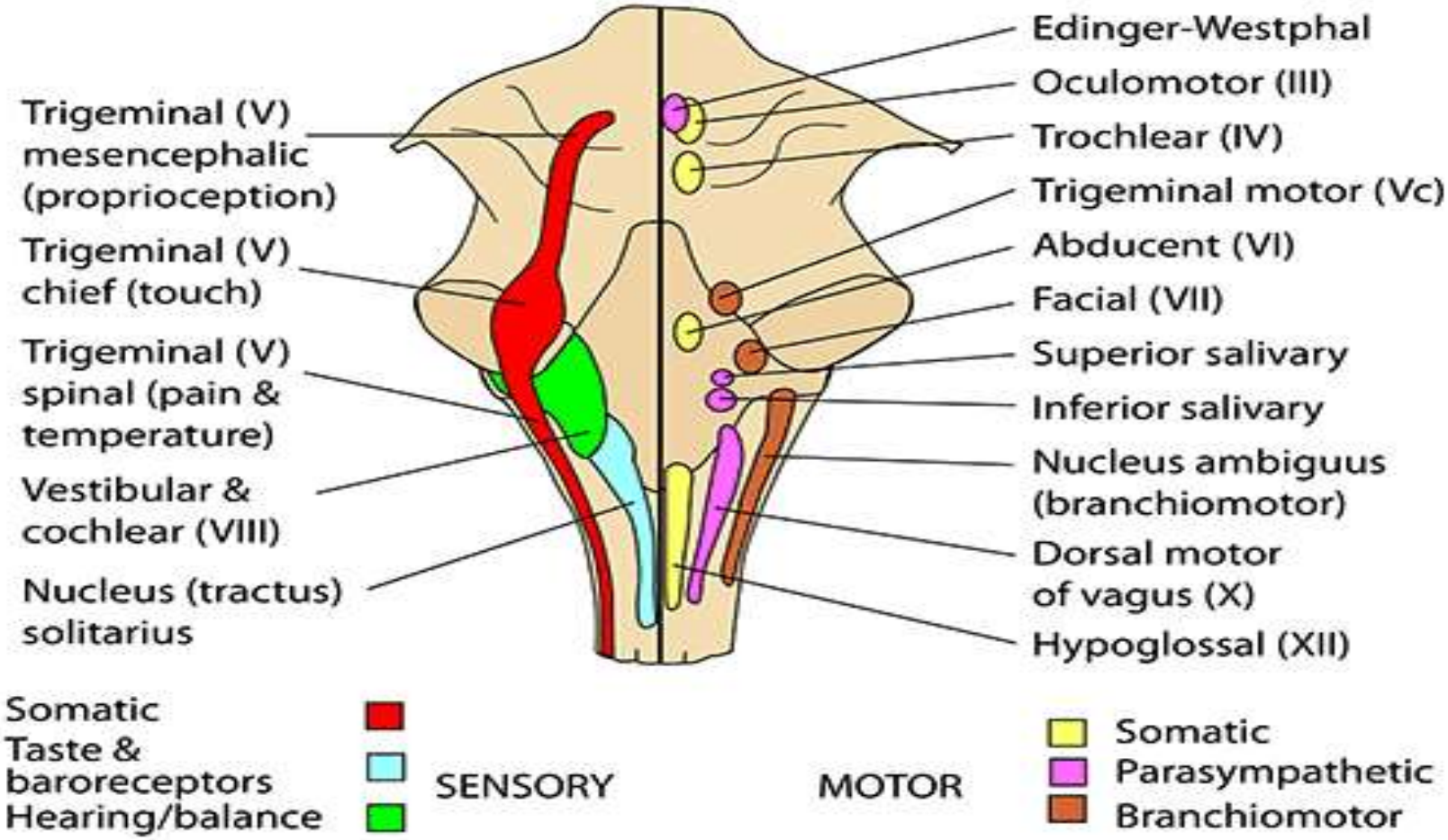


- **Nucleus ambiguus of 9th 10th & 11th CNs:**
  - Fibres from this nucleus represents a **special visceral efferent (SVE) cell column** whose axons innervate **pharyngeal arch muscles** of the **larynx** & **pharynx**.
  - These fibers contribute to parts of **9th 10th** & **11th CNs**
  - They exit the **medulla** via the **postolivary sulcus**.

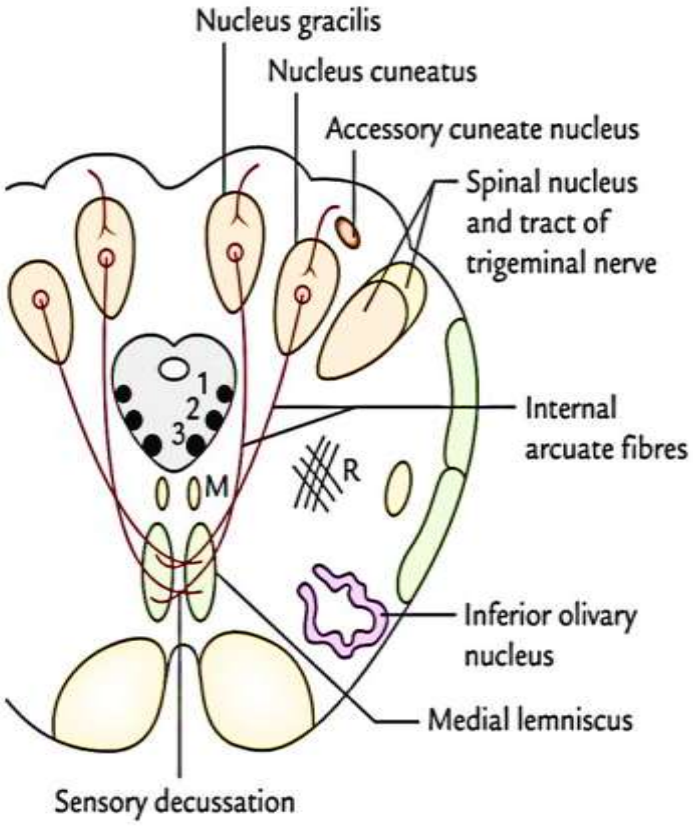




CRANIAL NERVE NUCLEI IN BRAIN STEM

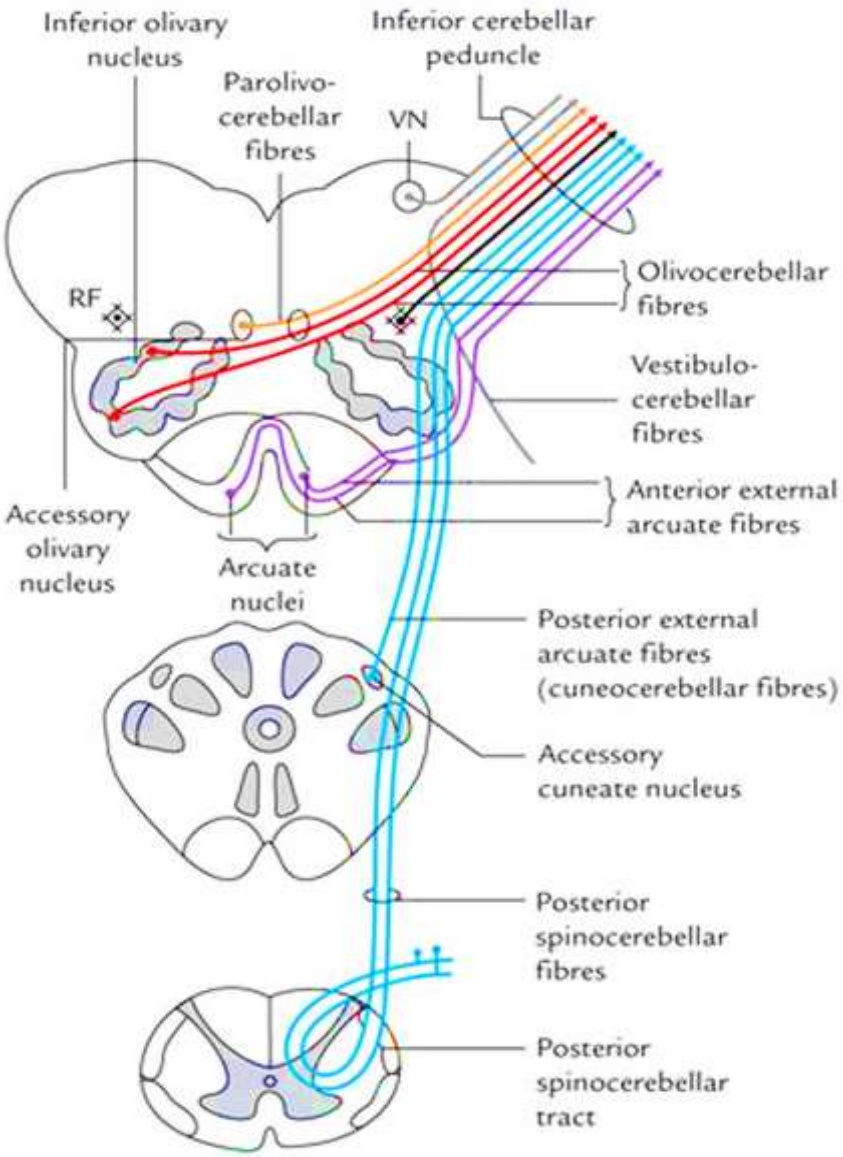


- **Nucleus of Tractus Solitarius (NTS)**
  - The nucleus receives **taste sensations from the tongue** via the **9th** & **10th CNs (chorda tympani branch of 7th CN** carries **taste sensation** also from **anterior 2/3rd** of the tongue**)**.
  - NTS lies lateral to the **dorsal nucleus of vagus**
- **Spinal Nucleus & Tract of Trigeminal Nerve:**
  - This **nucleus** (medial) & its **tract** (lateral) lies **close** to the **surface of the medulla oblongata**
  - The **nucleus** is in line with the **substantia gelatinosa** of the **spinal cord** while the tract with the **dorsolateral tract of Lissauer,** both are affected in **lesions** of the **lateral part of the medulla.**
  - The cells of the **spinal nucleus of trigeminal nerve** send their axons **(2nd order neurons)** to cross the **median plane** & **ascend** as the **trigeminal lemniscus** that **relays** in the **ventral nucleus of the thalamus**.
  - The **spinal tract** conveys **pain** & **thermal sensation** from the **face.**

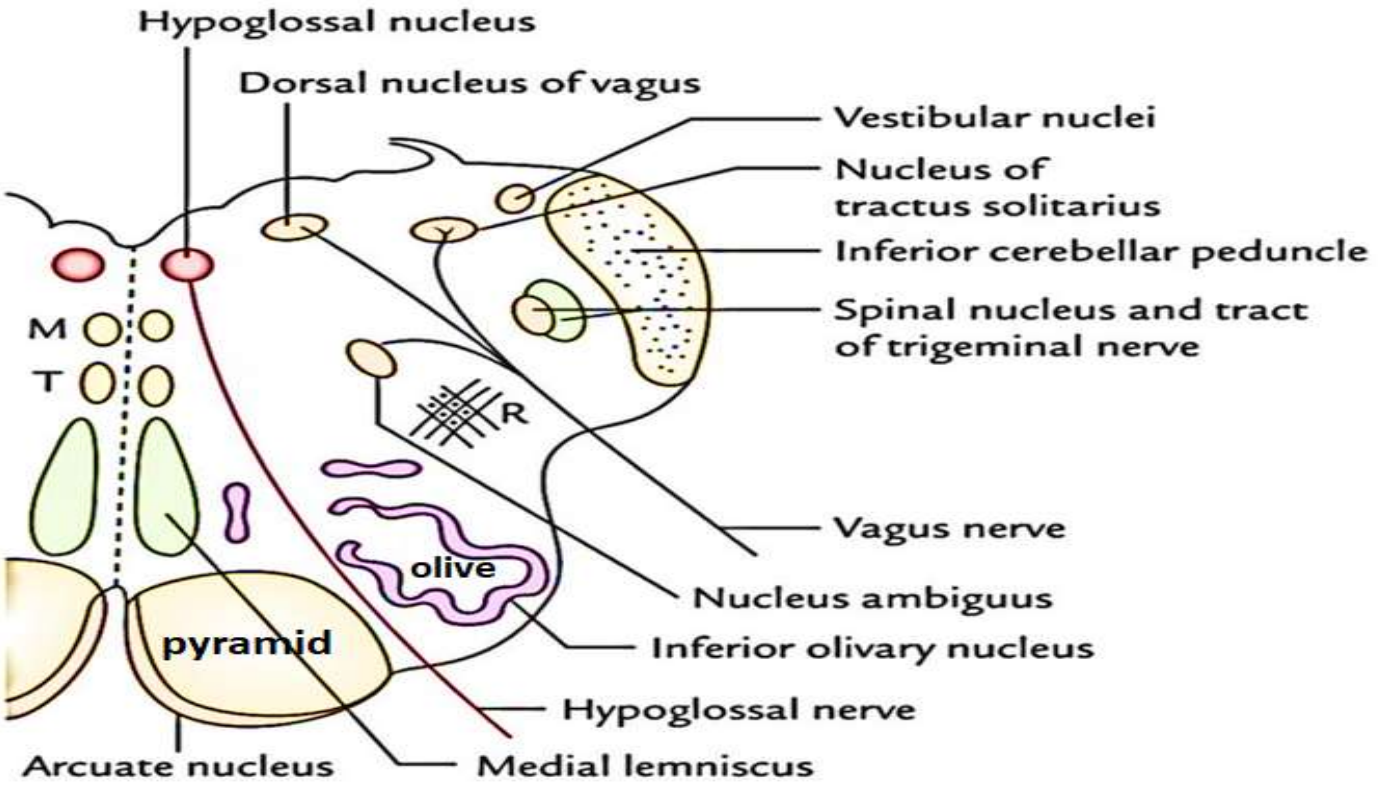


Inferior Olivary nucleus:

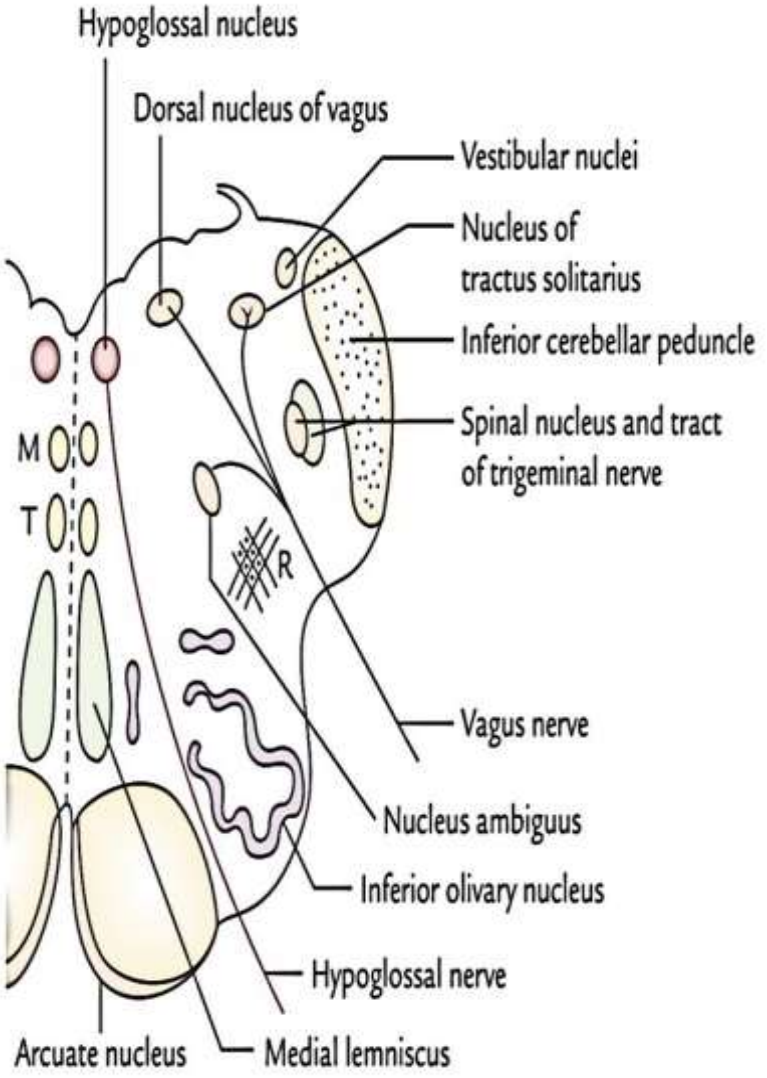
- The **inferior olivary nucleus** (ION) lie beneath the **olive**, receives **input** from the **red nucleus**
- ION is a large **hollow mass of gray matter crenated surface** (irregular notched edges) & **bulges on the surface** of **olive**.
- From the **hilus** of **ION** arises the **olivocerebellar fibers** which are its main **efferents** which **crosses** to the **opposite side** to enter the **cerebellum** through the **inferior cerebellar peduncle**.
- In other words, ION is a **cerebellar relay nucleus** that projects **olivocerebellar fibers** via the **inferior cerebellar peduncle** to the **contralateral cerebellar cortex** & **cerebellar nuclei**.



- **Vestibular Nuclei (Inferior & medial) of 8th CN:**
  - These **nuclei** receive **proprioceptive (special somatic afferent [SSA])** input from the **semicircular ducts, utricle, saccule & cerebellum** & project it to the **cerebellum** & **Medial Longitudinal Fasciculus (MLF**).



- **Arcuate nucleus:**
  - They are located on the **anterior surface** of the **medullary pyramids** as an extension of the **pontine nuclei**.
  - They receive fibers from the **corticospinal tract** & send their **axons** through the **anterior external arcuate fibers** & **medullary striae** to the **cerebellum** via the **inferior cerebellar peduncle**.
  - **Arcuate nuclei** are have a role in the **respiratory center** controlling the **breathing rate**.
- **Reticular formation nuclei:**
  - They are found **deep** within the **brainstem**, along its length. The **nuclei** can be **divided** into **three groups**. There is a **lateral**, a **medial** & a **median** (raphe) **group** of **nuclei**.

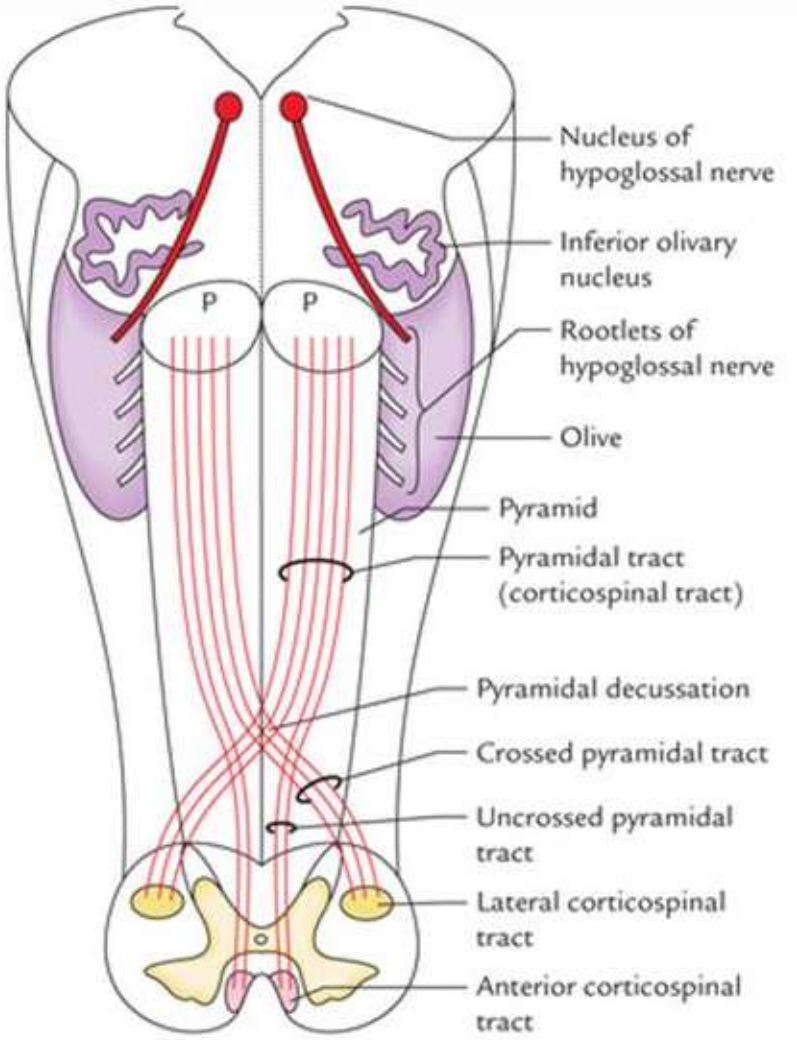


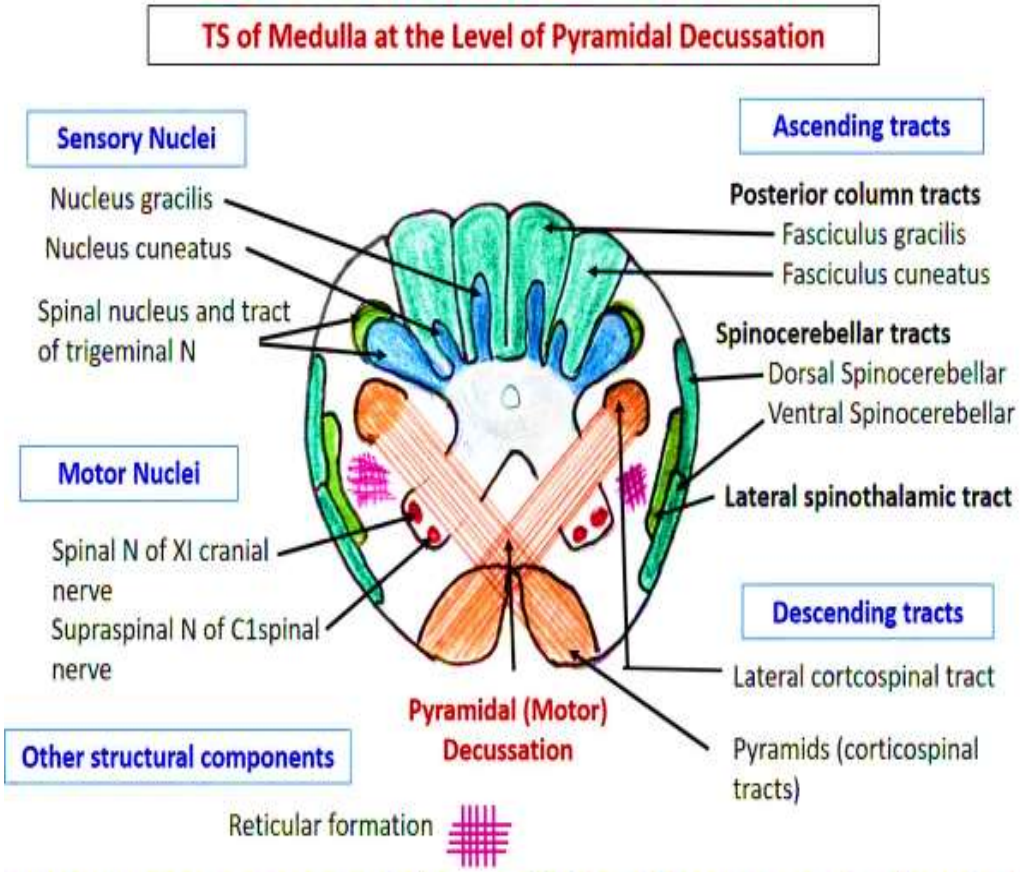
White Matter of Medulla Oblongata

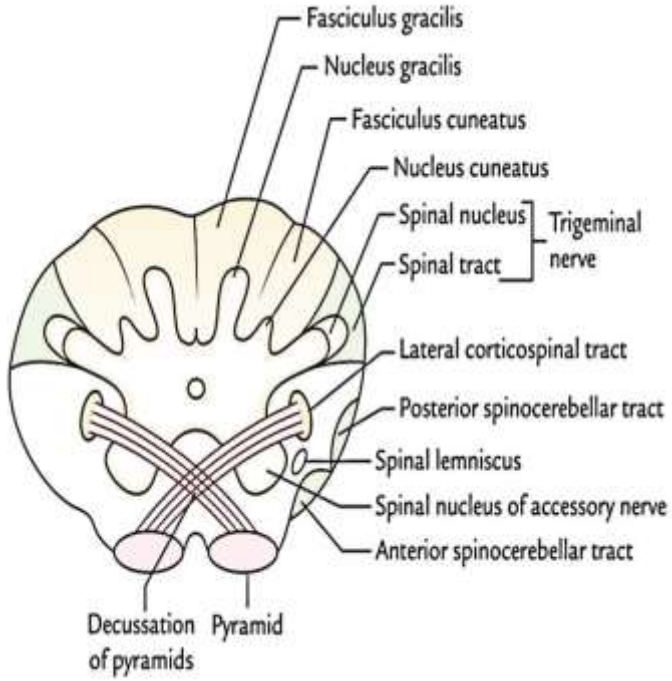
- The **Tracts** are gathered in **2 regions of the medulla:**
  - 1. Close to **Median plane:** These are (from afore backwards), the
    - (a) **pyramidal tract,**
    - (b) **medial lemniscus,**
    - (c) **tectospinal** &
    - (d) the **most posterior tract** throughout the whole brain stem the **medial longitudinal bundle (MLB)**.
  - 2. At the **Anterolateral region:** These are mainly the
    - (a) **posterior** &
    - (b) **anterior spinocerebellar tracts,**
    - (c) **lateral spinothalamic tract** &
    - (d) **spinal root of trigeminal nerve**.

Pyramidal Tract:

- It occupies the **pyramids** comprise the **corticobulbar fibers** & **corticospinal fibres**
- The **corticobulbar fibers** synapse with the **motor nuclei** of the **cranial nerves** of **opposite side**.
- About ¾ of the **corticospinal fibers** cross to the **opposite side** (called cross **pyramidal fibers**) in the **lowermost part of the medulla (pyramidal** or **motor decussation)**. The **remaining** ¼ of these fibers descends inside the **spinal cord** without **crossing** & forms the **anterior corticospinal tract (**also called **direct pyramidal fibers).**

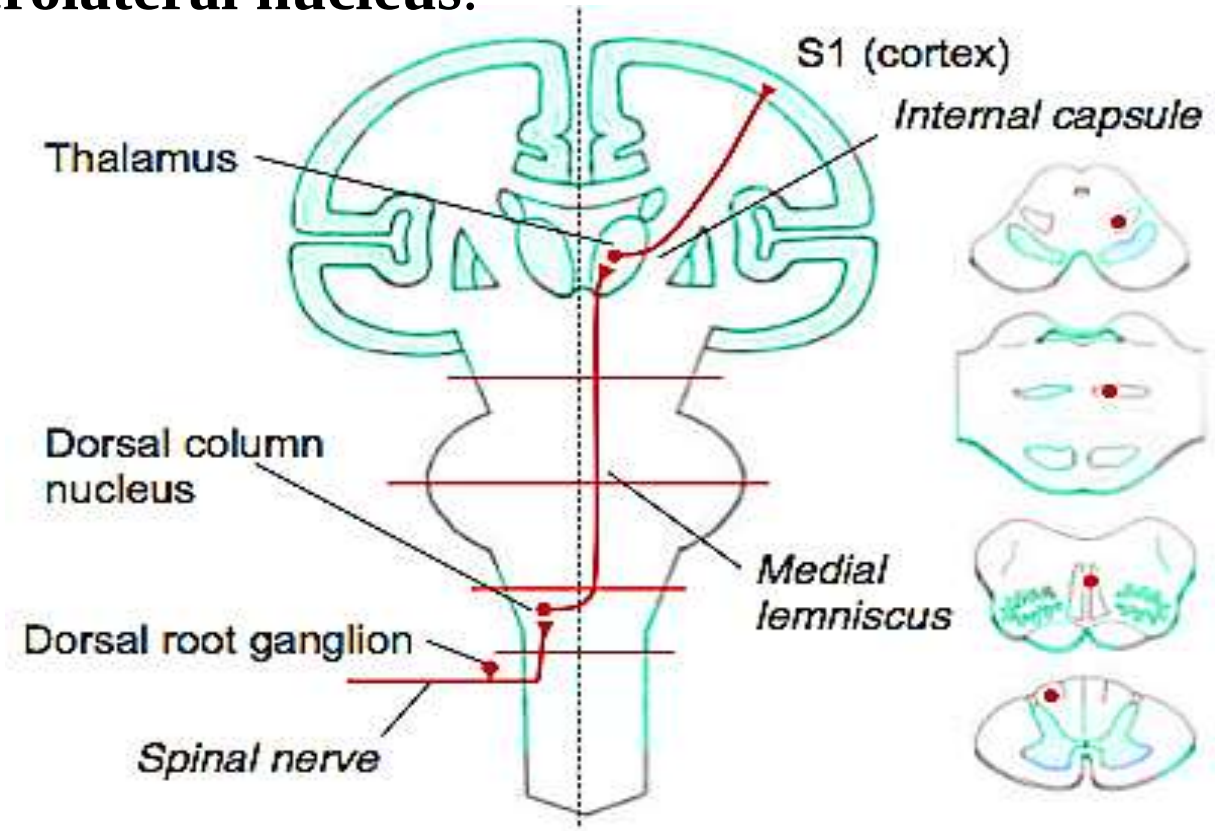




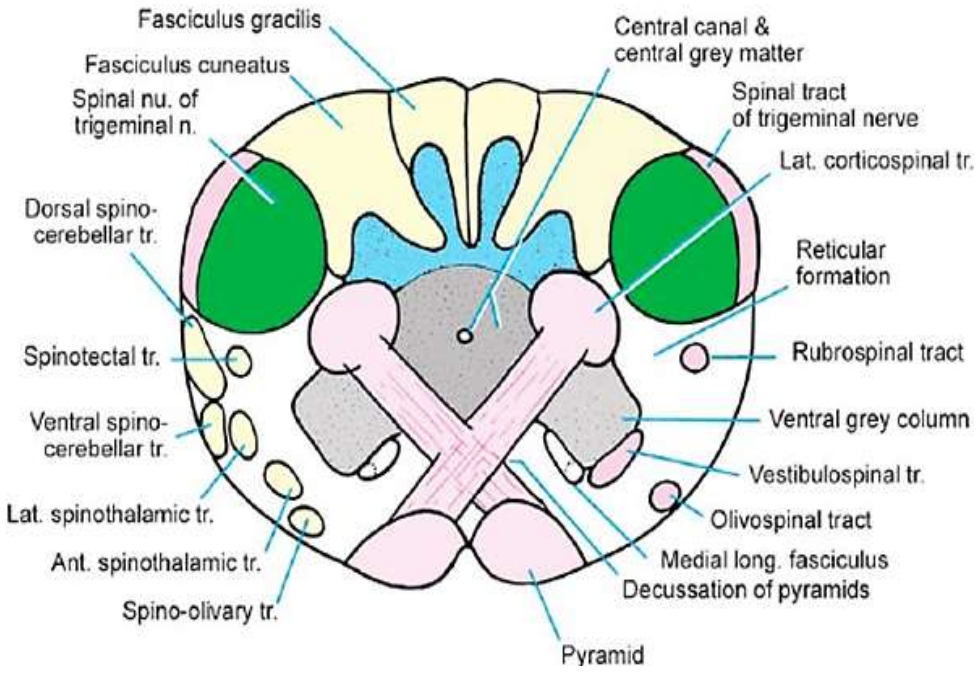


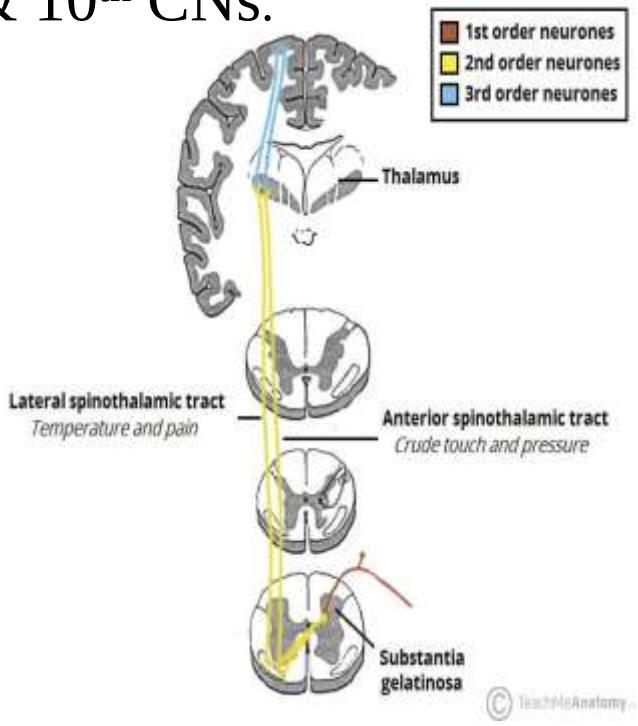
Transverse sections of the medulla oblongata at the level of pyramidal decussation.

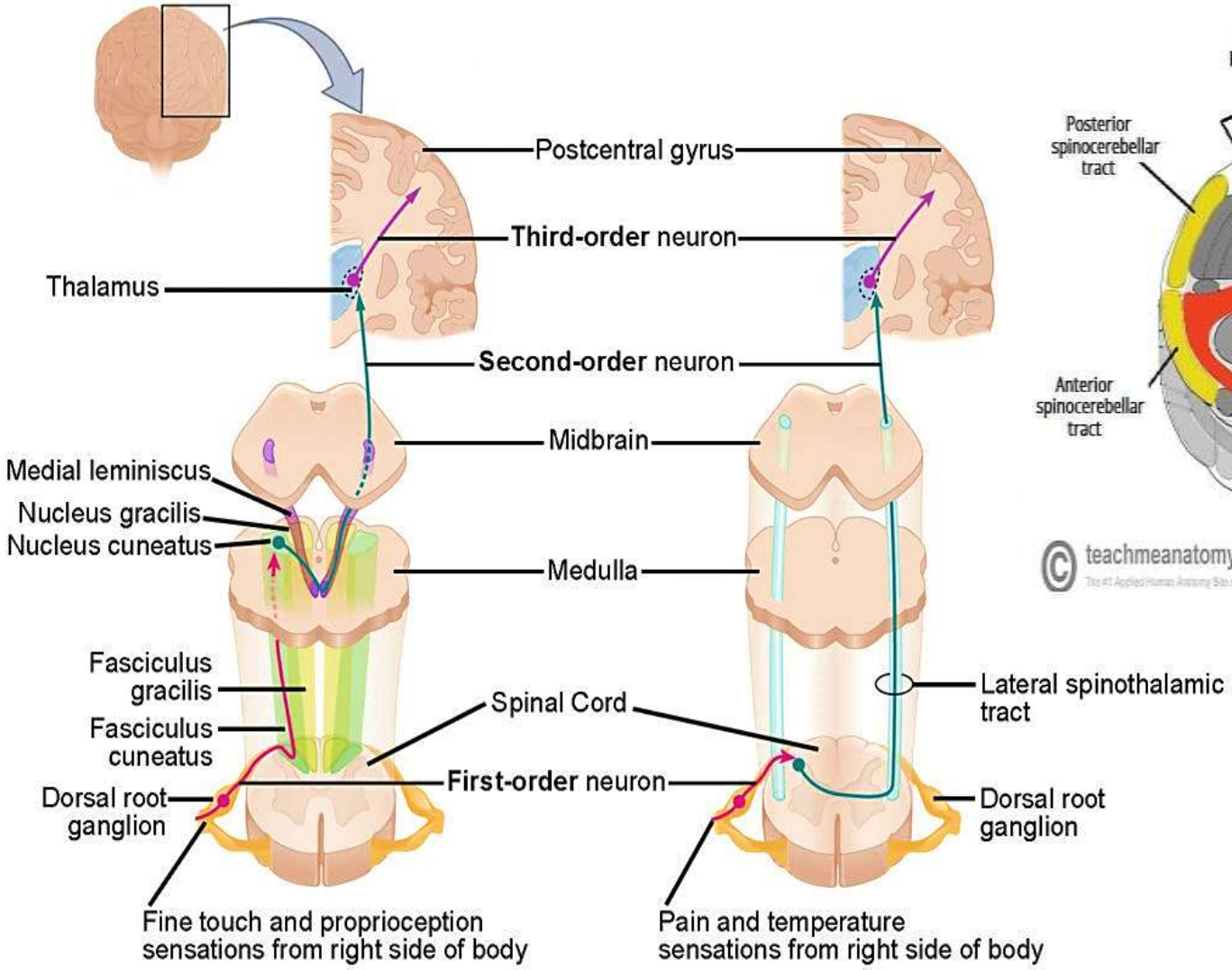
- **Medial lemniscus:**
  - It is a **flattened compact tract** formed after **decussation** of the **internal arcuate fibers**.
  - The fibers represent the axons of the **gracile** & **cuneate nuclei (i.e. 2nd order neurons).**
  - It conveys **dorsal column modalities** to the thalamic **ventral posterolateral nucleus**.

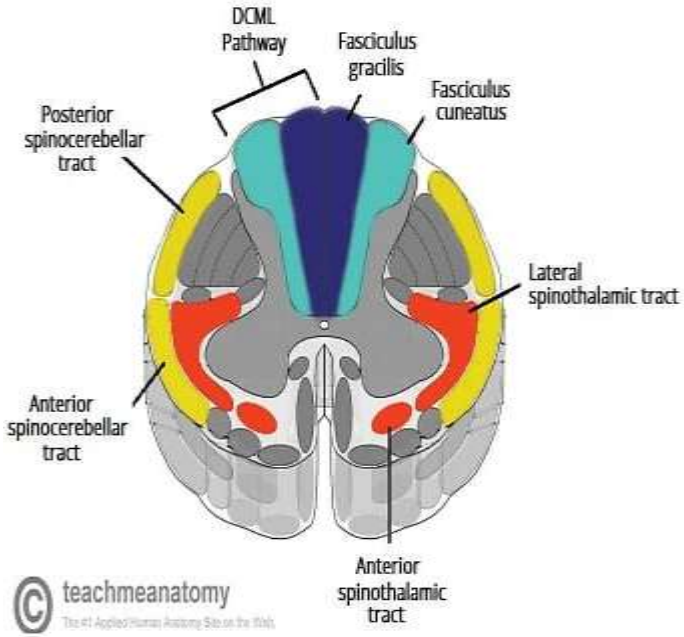


- **Spinal lemniscus:**
  - It contains the **spinothalamic (lateral** & **anterior) tracts** & the **spinotectal tract.**
  - It is the **upward continuation** of the **lateral spinothalamic tract** in the **brain stem** after taking the form of a **flattened compact band**.
  - It lies **posterior** to the **olivary nucleus** & is separated from the **lateral surface** by the **anterior spinocerebellar** & **spinotectal tracts**.
  - **Spinal tract of trigeminal nerve** replaces the **dorsolateral tract** of Lissauer. It contains **first-order neuron general somatic afferent (GSA) fibers** that mediate **pain**, **temperature & light touch sensations** from the **face** via 5th, 7th, 9th & 10th CNs.







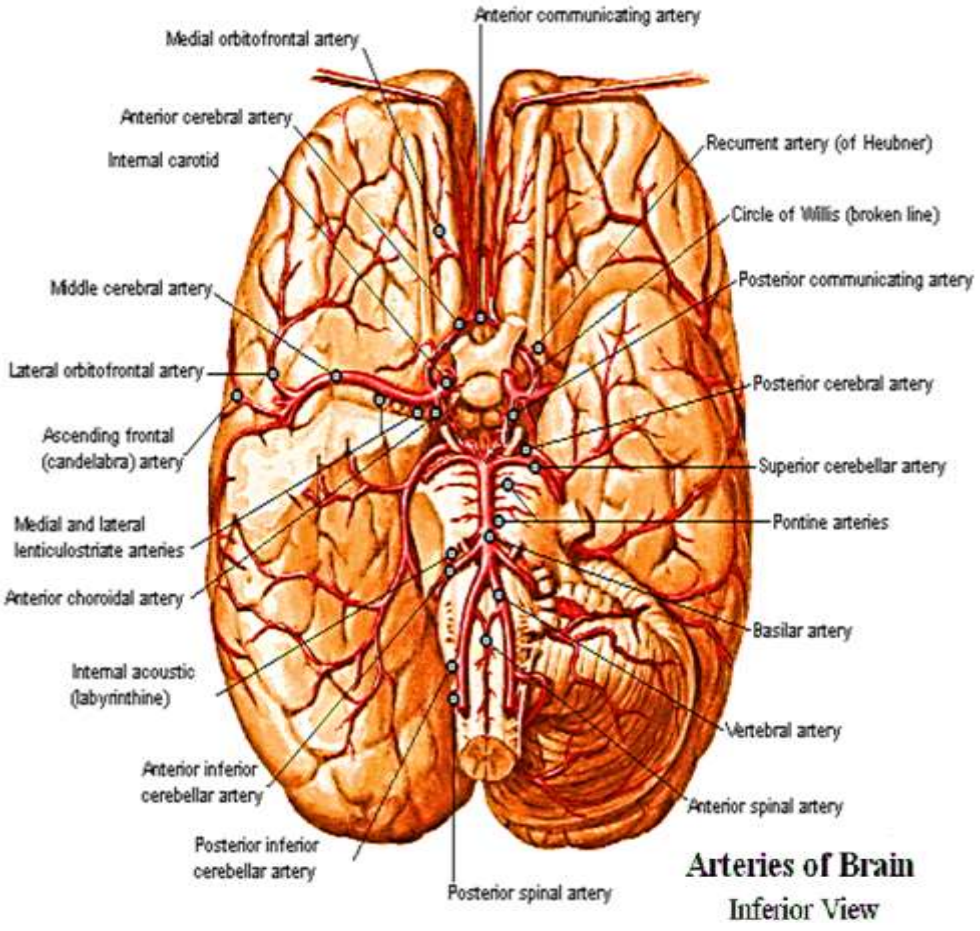


Dorsal column system

Spinothalamic tract

Blood Supply of the Medulla Oblongata

- The **medulla oblongata** is **supplied** by the:
  - **Vertebral arteries**.
  - **Spinal arteries** (anterior & posterior).
  - **Posterior inferior cerebellar arteries**.
  - **Basilar artery**.



Applied Anatomy

- Residing in the **medulla oblongata (MO)** are the **"vital centres"** comprising the **CRV: Cardiac Respiratory** & **Vasomotor centres.** Respiratory failure may arise from **compression** of the **respiratory center** is especially susceptible from injury or **poliomyelitis**.
- **Effect of increased intracranial pressure in the posterior cranial fossa to the MO**
- ***MO lies beneath the tentorium cerebelli & above the foramen magnum in the posterior cranial fossa, it is related anteriorly to the basal portion of the occipital bone & the upper part of the odontoid process of the axis & posteriorly to the cerebellum.***
- The intracranial pressure is elevated in **patients with tumors of posterior cranial fossa** & the brain (i.e. the cerebellum & MO) tends to be pushed toward the **area of least resistance**; resulting in a **downward herniation of the medulla & cerebellar tonsils through the foramen magnum.**
- **Due to traction**, this will cause **symptoms** of headache, neck stiffness & paralysis of the glossopharyngeal, vagus, accessory & hypoglossal nerves. In these **circumstances,** it is extremely **dangerous to perform a lumbar puncture** because the **sudden withdrawal** of **cerebrospinal fluid** may **precipitate further herniation of the brain** through the **foramen magnum** & abrupt **failure of vital functions**, resulting from **pressure** & **ischemia** of the **CN nuclei** present in **MO**.

- **Arnold-Chiari Phenomenon**
- The **Arnold-Chiari malformation** is a **congenital anomaly** in which there is a **herniation** of the **tonsils of the cerebellum** & the **medulla oblongata** via the **foramen magnum** into the **vertebral canal**.
- This results in **internal hydrocephalus** because of the **blockage** of the exits in the **roof of 4th ventricle** to the **cerebrospinal fluid**.
- It is commonly associated with **craniovertebral anomalies** or various forms of **spina bifida.**
- **Signs & symptoms** related to **pressure on the cerebellum & medulla oblongata** & involvement of the last **4 CNs** are associated with this condition.

Vascular Disorders of Medulla Oblongata

- **Lateral Medullary Syndrome of Wallenberg: Cause:** Lesion of the dorsolateral part of the medulla. ***As a result of the occlusion of the posterior inferior cerebellar artery (PICA)***
- **Affected Structures & Resultant Deficits Include:**

a) The **vestibular nuclei:** Lesions result in nystagmus, nausea, vomiting & vertigo.

b) The **inferior cerebellar peduncle:** Lesions will give rise to cerebellar signs e.g. past pointing, dysdiadochokinesia (dysmetria) & dystaxia in **ipsilateral** side.

c) The **nucleus ambiguus of 9th, 10th & 11th CNs:** Lesions will give rise to **hemiparalysis** of the **palates, pharynx** & **larynx** of the **ipsilateral** side i.e., loss of the gag reflex (efferent limb), dysarthria, dysphagia & hoarseness of voice.

d) The **glossopharyngeal nerve roots:** Lesions will cause **absence of the gag reflex** (afferent limb).

e) The **vagal nerve roots:** Similar to the lesions of the **nucleus ambiguus.**

f) The **spinal lemniscus (spinothalamic tracts):** Lesions will give rise to **loss of pain** & **temperature sensation** in the trunk & extremities of the **contralateral sides**.

g) The **spinal trigeminal nucleus & tract:** Lesions here will give rise to **loss of pain** & **temperature sensation** in the face of the **ipsilateral sides**.

h) The **descending sympathetic tract:** Lesions will cause an **ipsilateral Horner's syndrome** (i.e., ptosis, miosis, hemianhidrosis & enophthalmos).

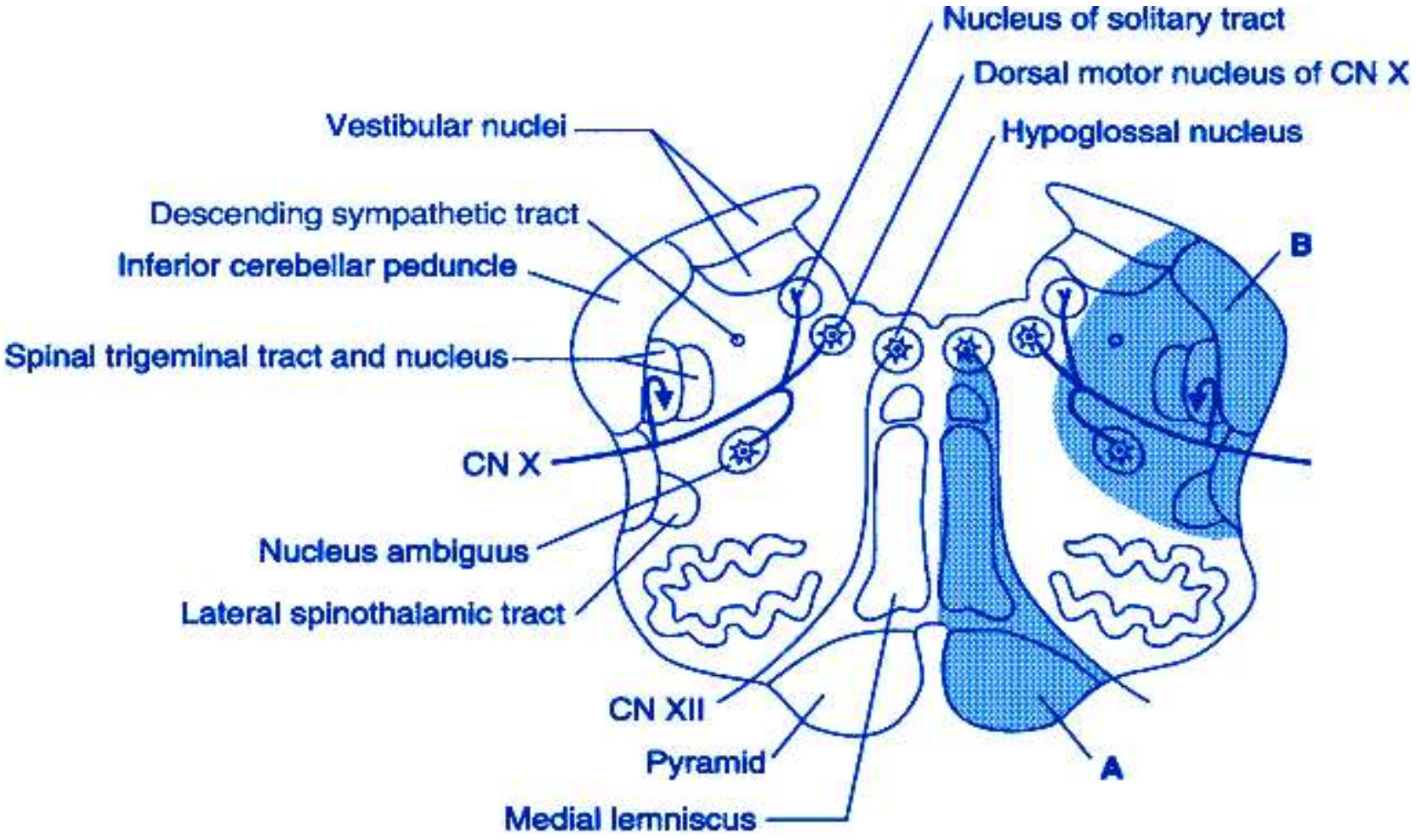
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- **Medial Medullary Syndrome: Cause:** Lesion of the ventral part of the medulla**.** ***As a result of the occlusion of anterior spinal artery***
- **Affected structures & resultant deficits include:**

a) The **corticospinal tract** (medullary pyramid): Lesions result in **contralateral spastic hemiparesis.**

b) The **medial lemniscus:** Lesions here will cause a **contralateral loss of tactile** & **vibration sensation** from the trunk & extremities.

c) The **hypoglossal nucleus (12th CN):** Lesions result in **ipsilateral flaccid hemiparalysis** of the tongue. When the tongue is protruded, it usually deflects to the side of weakness which is the **side of the lesion**. Often produces the lesion commonly called ***Alternating hypoglossal hemiplegia***


Nucleus of solitary tract
Dorsal motor nucleus of CN X
Hypoglossal nucleus
Vestibular nuclei
Descending sympathetic tract
Inferior cerebellar peduncle
Spinal trigeminal tract and nucleus
CN X
Nucleus ambiguus
Lateral spinothalamic tract
CN XII
Pyramid
Medial lemniscus
B
A

